Optimal Government Debt in an Open Economy*

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Abstract

I study the optimal level of government debt in a stylized heterogeneous agents open economy model. First, I show that a Ramsey planner faces a key tradeoff between providing liquidity to domestic agents and manipulating interest rates to maximize monopoly rents when selling the debt to foreign agents. Second, I derive necessary conditions to characterize the optimal level of debt and interest rates. The planner optimally sets lower interest rates compared to the closed economy first best in order to reduce interest payments to foreign agents. This tradeoff intensifies for more inelastic foreign agents' demand for domestic debt. Finally, I show that the planner can achieve the first best allocation by issuing two separate debt instruments for domestic and foreign agents.

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1 Introduction

The *post* Great Recession world has witnessed large increases in debt burdens across advanced economies. For instance, the U.S. debt-to-GDP ratio, has surged from 67% in 2008 to approximately 121% in 2024. Simultaneously, government debt markets have become increasingly integrated, with foreign investors now holding substantial portions of domestic debt (e.g., the average share of U.S. government debt held by foreigners between 2008 and 2024 is 29%).

These two aspects – rising debt levels and international financial integration – create an environment where domestic and international considerations interact in determining optimal debt policy. For instance, governments may optimally issue debt as a tool to provide liquidity and collateral to the domestic economy. At the same time, in open economies, governments face incentives to manage capital flows and interest rates, decisions that indeed affect the optimal issuance of sovereign debt.

This paper investigates the optimal level of government debt in an open economy, bridging two distinct strands of the macroeconomic and international literature. (I) The first strand examines government debt in closed economies with heterogeneous agents and incomplete markets. In particular, government debt provides private liquidity and insurance in heterogeneous agents incomplete markets model with idiosyncratic risk (e.g., Woodford, 1990; Aiyagari and McGrattan, 1998). In this setting, where households want to hold precautionary savings, debt represents both a saving vehicle for agents and a redistributive tool from workers to savers (through higher interest rates). (II) The second strand of the literature explores government debt from an open economy perspective, focusing primarily on capital controls and safe asset provision. In particular, a domestic planner has incentives to manipulate interest rates, distorting global prices to extract monopoly rents from foreign countries (Costinot, Lorenzoni, and Werning, 2014). Furthermore, a country issuing debt as a global safe asset tradeoffs extraction of seigniorage revenues from foreign investors against increased default risk that accompanies excessive borrowing (Farhi and Maggiori, 2018). Hence, a government selling debt to foreign residents can strategically behave as a monopolist, manipulating interest rates to extract rents from international investors.

Overall, domestic liquidity provision and international rents extraction have been studied separately in the literature. My work instead shows how these two motives interact to determine the optimal level of government debt.

This paper's primary contribution lies in demonstrating that the two policy objectives of domestic liquidity provision and international rents extraction generate an inherent and economically significant trade-off. The government must balance competing targets: (1) providing liquidity and insurance to domestic households, and (2) extracting monopoly rents when selling debt internationally. These objectives typically require contradictory policy actions, creating a fundamental tension in optimal debt management. In fact, domestic insurance provision typically requires expanding the debt supply, with higher interest rates, while rent extraction from foreign investors requires reducing borrowing to lower interest rates. This tension is particularly acute for economies facing an inelastic foreign demand for their debt, as their potential for interest rate manipulation – thus rent extraction – is greater. The following analysis characterizes how a welfare-maximizing planner should set debt levels to optimally navigate this trade-off and suggests a mechanism to decentralize the first-best allocation through separate domestic and foreign debt instruments.

First, I establish this trade-off formally, considering a fully-insured economy with a representative agent framework. When a Ramsey planner can issue debt exclusively to the rest of the

world (RoW) while preventing domestic agents from purchasing it, the planner optimally reduces debt issuance to lower interest rates and gain on infra-marginal units of debt. The more inelastic the foreign demand, the greater the interest rate reduction. This scenario describes the first-best outcome, where the domestic economy is fully insured and interest rates are optimally manipulated.

However, in the more realistic scenario where the planner cannot prevent domestic agents from purchasing government debt, its ability to control interest rates vanishes. Through a Ricardian equivalence mechanism, debt paths no longer affect equilibrium allocations or prices. Domestic agents, who price the debt through the standard Euler equation, fail to internalize how their saving decisions influence interest rates. Consequently, equilibrium interest rates exceed first-best levels, and the economy over-borrows from international markets. This demonstrates that without the ability to discriminate between domestic and foreign debt purchasers, achieving the first-best allocation becomes impossible. Therefore, the planner could adopt second-best solutions by deviating from full domestic insurance to regain the ability to set interest rates optimally.

Second, I examine the key trade-off between domestic liquidity provision and international rents extraction within a stylized heterogeneous agents framework based on Woodford (1990). The model features two agents with alternating high and low endowments across periods and zero-borrowing constraints: an agent who is rich (has a high endowment) in a given period will be poor in the following one. In this setting, the credit constraint prevents a poor agent from borrowing against her higher future income to smooth consumption. This framework is analytically tractable and captures key features of more general Bewley models. In particular, a subset of agents operates off their Euler equations with high marginal propensity to consume (MPC). Moreover, Ricardian equivalence fails, allowing the planner to influence interest rates through debt issuance (in general, the equilibrium interest rate is increasing in the level of debt). Within this framework, I study the optimal debt policy of a Ramsey planner maximizing utilitarian welfare and issuing debt under full commitment to repay it. The planner is not able to discriminate between domestic and foreign agents when selling the debt.

Initially, I build intuition with a two-period version of the model. The planner's first-order condition reveals two competing forces. (i) *Partial insurance*, defined as the wedge in the Euler equation of poor agents (who are liquidity constrained). This force gives the planner an incentive to increase debt levels to provide liquidity to smooth poor agents' consumption over time. (ii) *Interest rate manipulation*, defined as the wedge in the Euler equation of domestic rich asset holders, who price the debt but do not internalize the effect of their savings decisions on the equilibrium interest rate. This force incentivizes debt reduction to reduce interest rates extracting monopoly rents from foreign investors. The magnitude of the latter effect increases with the inelasticity of foreign demand. Overall, the planner will only insure partially the economy to extract some monopoly rents from the RoW.

Then, I extend the model to the infinite horizon. The Ramsey planner optimally selects the sequence of interest rates and tax rates, while the path of debt is determined from the economy implementability constraints. Focusing on the Ramsey steady state (RSS) – defined as limit of the Ramsey plan (the sequence of interest and tax rates that solve the Ramsey problem) – I derive a necessary condition that links tax rates and interest rates to RoW demand for domestic debt. The planner optimally choses to partially insure the economy, hence reducing interest rates to extract international rents.

I also show that this partial insurance result holds if the planner discounts future utilities differently from the households. In this case, optimal policy depends on the *net liquidity benefits from the RoW* when issuing debt. In fact, a higher interest rate increases foreign borrowing, with

a short-term marginal benefit from relaxing the country budget constraints but with a long-term marginal cost from higher interest payments. A more patient planner, who places greater weight on long-run costs, reduces interest rates more aggressively to limit foreign borrowing. Furthermore, the elasticity of RoW demand interacts with the net liquidity benefits, depending on how much the planner weights long-run outcomes relative to the short-run.

Finally, I show that in the infinite horizon model, the planner can achieve the first best of full domestic insurance and optimal interest rate manipulation (or international rents extraction) by issuing two separate debt instruments: domestic debt (purchasable only by domestic agents) and foreign debt (restricted to international investors). With this dual-instrument approach, the planner optimally sets the tax rate and distinct interest rates on domestic and foreign debt. Following Tinbergen's principle that optimal policy requires matching instruments to targets, this approach enables simultaneous achievement of domestic insurance and international rents extraction.

The paper proceeds as follows. Section 2 establishes the fundamental trade-off between domestic insurance and interest rate manipulation. Section 3 analyzes this trade-off in a two-period heterogeneous agent framework. Section 4 extends the analysis to the infinite horizon. Section 5 demonstrates how separate debt instruments can decentralize the first-best allocation. Section 6 concludes.

1.1 Literature review

I relate to the literature on optimal government debt in the closed economy and on the optimal provision of safe assets and capital controls in open economies.

Closed economy. Early work on optimal government debt dates back to the 60s and 70s. Diamond (1965) studies optimal government debt in a neoclassical growth model with overlapping generations. Both external and internal debt affect the long-run competitive equilibrium through higher tax rates and a crowding-out effect on physical capital (for internal debt). Barro (1979) shows that if Ricardian equivalence holds but there are second-order costs from taxation, the government optimally chooses the path of debt to smooth taxes over time. The level of debt is however determined only by initial conditions.

Following work has focused on the determinants of the *level* of optimal government debt in heterogeneous agents economies. Woodford (1990) examines an analytical two-agent model with alternating high and low endowments across periods. Higher debt increases the steady-state interest rate providing liquidity to low-endowment agents. In this paper, I study an open-economy version of the Woodford model. Aiyagari and McGrattan (1998) evaluate the optimal steady-state level of debt in an Aiyagari (1994) economy with physical capital and labor taxes to finance the debt. Optimal debt balances the benefit of providing liquidity to household with redistribution costs and crowding-out of physical capital. Holmström and Tirole (1998) focus on the firms side, as government debt provides liquidity for firms (in a setting where liquidity shocks and moral hazard induce a demand for liquidity). Overall, I borrow from this literature the notion that debt is beneficial to the economy since it provides liquidity and insurance.

More recently, Angeletos, Collard, and Dellas (2023) have studied optimal debt policy (both in the short-run and long-run) in a context where taxes are distortionary and debt provides some social benefits (e.g., liquidity provision, easing financial frictions). The planner balances three forces: smoothing tax distortions over time, providing social benefits from debt issuance, and

maximizing seignorage revenues (in terms of low interest rates on debt). Differently from them, I abstract from tax distortions, but additionally study the incentive to extract seignorage revenues from foreign holders of the debt. Le Grand and Ragot (2025) study optimal labor and capital tax and debt policy in Aiyagari-type economies and characterize the steady state of Ramsey plans. They first analyze the Woodford (1990) model with general utility functions, then consider a more quantitative Aiyagari model. While I also focus on the Woodford model, I abstract from capital taxes and adopt log-utility. On the contrary I expand their analysis to an open economy.

Auclert, Cai, Rognlie, and Straub (WP) study the Ramsey problem in Aiyagari models, using a sequence-space representation. They characterize the steady state of the Ramsey plan through necessary conditions that involve infinitely discounted elasticity of asset and labor aggregate policy functions to prices. They show that standard Aiyagari models are subject to an immiseration result since the Ramsey steady state either does not exist or features high labor taxes and low consumption. They also show that immiseration holds in a two-country open Aiyagari economy model. I follow their approach to characterize Ramsey steady states through necessary conditions, but I focus on an analytical treatment of the trade-off between liquidity provision and interest rates manipulation.

Other works have studied optimal policy with a focus on inflation tax and on the issues of time-consistency and commitment (e.g., Lucas and Stokey, 1983; Chari and Kehoe, 1999, Debortoli et al., 2021; Sims, forthcoming). I abstract from these issues here.

Open economy. The international macroeconomics literature has studied capital controls and the optimal provision of safe assets.

Costinot, Lorenzoni, and Werning (2014) consider a two-country open economy model where the domestic planner manipulates dynamic terms-of-trade. When the home country is growing faster it is optimal to tax capital inflows to promote domestic savings. In fact, as output is higher in the future, net exports are larger, hence the country has an incentive to increase the prices of future goods (to gain on infra-marginal units of net exports). This is achieved by higher future consumption (hence larger current savings). Farhi and Maggiori (2018) examine the optimal supply of a safe asset from an hegemon country, that obtains monopoly rents in the form of a positive endogenous safety premium on the asset. However, excessive issuance may lead to confidence crisis and default. The country balances these two motives when issuing the safe asset. Choi, Kirpalani, and Perez (2024) additionally show in the data that the U.S. behaves as a monopolist when issuing its debt, exploiting its market power and significantly under-providing the safe asset.[1] I connect to this literature since I study a planner that has an incentive to manipulate interest rates to extract monopoly rents from abroad. Differently from these papers, I consider an heterogeneous agents model, additionally taking into account domestic liquidity and insurance provision benefits that arise from government debt issuance. I also abstract from default risk, assuming that the planner fully commits to repay the debt.

Other papers have focused on optimal tax and debt policy in a small open economy. In particular, Anderson (1991) and Anderson and Young (1992) study borrowing and consumption taxes in a representative agent economy, together with the correlation properties of public and private deficits and current account. Differently from them, I consider agents heterogeneity and abstract from capital. Aguiar, Amador, and Gopinath (2009) examine optimal (credible) taxation on foreign capital and sovereign debt policies in a small open economy, exploring the link between debt and

[1] Schmitt-Grohé and Uribe (2012) also find that a planner has an incentive to extract seignorage from foreigners, but focus on optimal inflation rate and deviations from the Friedman rule.

investment. Finally, Mendoza, Quadrini, and Ríos-Rull (2009) and Caballero, Farhi, and Gourinchas (2008) study models of global imbalances where some countries have an excessive supply of savings (either due to more incomplete markets or inability to generate sufficient local store of value instruments).

2 Liquidity provision and interest rate manipulation

In this section I show that a Ramsey planner that wishes to achieve both full domestic insurance and optimal monopoly rents extraction from foreigners faces a trade-off between these two targets. In particular, my argument is based on showing that if a planner keeps the economy at full domestic insurance, then it loses the ability to manipulate interest rates and extract rents from the abroad provided that it is not able to discriminate between domestic and foreign holders of domestic debt.

2.1 Model Setup

I assume that the planner keeps the economy at full domestic insurance. This means that the planner is able to complete the domestic market by allowing domestic agents to trade Arrow securities and relaxing any borrowing constraint. Under complete markets, it is possible to aggregate individual preferences and describe the domestic economy with a representative agent.

Representative agent. I consider here a representative agent whose preferences are described by a time-separable lifetime utility. Time t utility function is $U(C_t)$ where C_t denotes aggregate time t consumption. I assume that $U(\cdot)$ is strictly increasing and concave and continuously differentiable. The agent discounts future utility with a discount factor $\beta \in (0,1)$. Time is discrete. In every period the agent receives an endowment Y_t. I assume that there is no aggregate risk.

Rest of the World. I assume that the economy is open in the sense that the rest of the world (RoW) can save in domestic assets. In particular, I model the RoW in a reduced form way, simply postulating that in every period t the RoW demands $A_t^*(r_{t+1})$ risk-free domestic assets. The risk-free real interest rate paid on the domestic asset between t and $t+1$ is r_{t+1}. I assume that the demand for the domestic asset is differentiable and increasing in the real interest rate, hence $\frac{\partial A_t^*}{\partial r_{t+1}} > 0$. Notice that I have implicitly assumed that foreign demand only depends on the interest rate between t and $t+1$.[2]

Ramsey planner. The government issues in every period t risk-free one-period government debt B_t paying the real interest rate r_{t+1} between t and $t+1$. The government chooses B_t optimally in every period behaving as a Ramsey planner that maximizes the representative agent's lifetime utility:

$$\sum_{t=0}^{\infty} \beta^t U(C_t).$$

I assume that the planner maintains full commitment to repay debt. The planner operates under the country aggregate resource constraint:

$$C_t = Y_t + A_t^* - (1 + r_t)A_{t-1}^* \tag{1}$$

[2]This is an analytically convenient assumption, while a simplification, since foreign bonds demand may depend on the entire sequence of interest rates.

that establishes that in every period aggregate consumption is equal to the endowment and the change in RoW holdings of domestic debt $A_t^* - A_{t-1}^*$, including interest payments to foreigners $-r_t A_{t-1}^*$. There is also an initial condition on debt B_{-1}.

I analyze three distinct environments, characterized by different assumptions about the planner's ability to restrict domestic bond purchases and whether the planner internalizes the effect of debt issuance on interest rates. In each of the three cases the planner faces different sets of constraints.

In particular I will study the planner's problem in the case in which it is able to discriminate between domestic and foreign agents in the sense of being able to restrict domestic agents from buying the debt. In such an environment the planner may or may not internalize the interest rate schedule as a function of the level of debt issuance (respectively Case 1 and Case 2). Lastly, I will show that if the planner is not able to prevent domestic agents from buying the bonds, then it will lose the ability to extract rents from foreign agents (Case 3).

2.2 Case 1: Restricted Domestic Demand without Interest Rate Internalization

In our first scenario, the planner restricts domestic agents from purchasing government bonds but does not internalize how debt issuance affects interest rates. Under these conditions, bond market clearing requires $B_t = A_t^*$, as only foreign investors can hold domestic debt. The equilibrium interest rate is given by the inverse of RoW demand. Let $r_{t+1} = \mathcal{R}_t(B_t)$ denote the interest rate schedule as a function of debt, then $A_t^*(\mathcal{R}_t(B_t)) = B_t$. Importantly, in this environment the planner treats r_{t+1} as exogenous when issuing debt. The planner's optimization problem is:

$$\begin{aligned} \max_{\{C_t, B_t\}} \quad & \sum_{t=0}^{\infty} \beta^t U(C_t) \\ \text{s.t.} \quad & C_t = Y_t + B_t - (1 + r_t) B_{t-1}. \end{aligned}$$

The first-order condition yields:

$$U'(C_t) = \beta(1 + r_{t+1}) U'(C_{t+1}). \tag{2}$$

This is the standard Euler equation, indicating that the planner equalizes the marginal utility of consumption across periods, adjusted for discounting and the interest rate. Since the planner does not internalize how debt issuance affects interest rates, the solution does not account for the price impact of additional borrowing. The planner thus issues debt until the marginal utility of consumption today equals the discounted marginal utility of consumption tomorrow, weighted by the gross interest rate.

Substituting the resource constraints in equation (2), together with the equilibrium interest rate schedule, I can characterize the necessary and sufficient conditions for the optimal path of debt:

$$\begin{aligned} & 1 + r_{t+1} = \frac{1}{\beta} \frac{U'(C_t)}{U'(C_{t+1})} = \frac{1}{\beta} \frac{U'\Big(Y_t + B_t - (1 + r_t) B_{t-1}\Big)}{U'\Big(Y_{t+1} + B_{t+1} - (1 + r_{t+1}) B_t\Big)}, \\ & r_{t+1} = \mathcal{R}_t(B_t). \end{aligned}$$

Finally, the optimal plan must also satisfy a transversality condition:

$$\lim_{T \to \infty} \beta^T U'(C_T) B_T = 0.$$

2.3 Case 2: Restricted Domestic Demand with Interest Rate Internalization

In our second case, the planner still restricts domestic agents from purchasing government bonds but now internalizes how debt issuance affects interest rates. The key difference is that the planner explicitly accounts for the relationship $r_{t+1} = \mathcal{R}_t(B_t)$ in the optimization process. This recognition of the interest rate response to debt issuance alters the planner's decision-making process. The planner's problem becomes:

$$\begin{aligned} \max_{\{C_t, B_t\}} \quad & \sum_{t=0}^{\infty} \beta^t U(C_t) \\ \text{s.t.} \quad & C_t = Y_t + B_t - (1 + r_t) B_{t-1}, \\ & r_{t+1} = \mathcal{R}_t(B_t). \end{aligned}$$

The first-order condition now yields:

$$U'(C_t) = \beta(1 + r_{t+1}) U'(C_{t+1})[1 + \Gamma_t] \tag{3}$$

where Γ_t represents the elasticity of the gross interest rate to debt:

$$\Gamma_t := \frac{\partial \mathcal{R}_t(B_t)}{\partial \log B_t} \frac{1}{1 + r_{t+1}}.$$

Since $A_t^* = B_t$, the elasticity Γ_t equals the inverse elasticity of RoW demand A_t^* and captures how much the interest rate increases when debt issuance rises. As foreign demand becomes more elastic, Γ_t decreases, reflecting a smaller interest rate response to changes in debt issuance.

Overall, the optimal path of debt is determined by equation (3), the resource constraints, and the transversality condition:[3]

$$\begin{aligned} (1 + r_{t+1})[1 + \Gamma_t(B_t)] &= \frac{1}{\beta} \frac{U'\Big(Y_t + B_t - (1 + r_t) B_{t-1}\Big)}{U'\Big(Y_{t+1} + B_{t+1} - (1 + r_{t+1}) B_t\Big)}, \\ r_{t+1} &= \mathcal{R}_t(B_t), \end{aligned}$$

where I have explicitly expressed the interest rate elasticity as a function of debt ($\Gamma_t(B_t)$).

Interpretation. According to the modified Euler equation (3), when the planner issues an additional unit of debt, it increases current consumption, with a welfare improvement $U'(C_t)$. However, in $t+1$ the planner repays (i) $1 + r_{t+1}$ on the additional unit of debt and (ii) higher interests on all the infra-marginal units of debt $B_t \cdot \partial r_{t+1} / \partial B_t$ since it internalizes that the interest rate is increasing in debt. Overall the marginal cost on debt issuance is given by the sum of (i) and (ii):

$$\beta U'(C_{t+1}) \left[1 + r_{t+1} + B_t \cdot \frac{\partial r_{t+1}}{\partial B_t} \right].$$

In other words, the planner internalizes the fact that higher debt issuance puts upward pressure on the interest rate. At the margin, the shadow price that the planner faces is $1 + r_{t+1} + B_t \cdot \frac{\partial r_{t+1}}{\partial B_t}$.

[3] The transversality condition is:

$$\lim_{T \to \infty} \beta^T U'(C_T) B_T = 0.$$

At the optimum, the marginal rate of substitution between current and future consumption equals this shadow price:

$$\frac{U'(C_t)}{\beta U'(C_{t+1})} = 1 + r_{t+1} + B_t \cdot \frac{\partial r_{t+1}}{\partial B_t} = (1 + r_{t+1})(1 + \Gamma_t)$$

Moreover, the optimality condition (3) implies that $U'(C_t) > \beta(1 + r_{t+1})U'(C_{t+1})$ since $\Gamma_t > 0$. This inequality reveals a key insight: the planner issues less debt compared to Case 1 (where it does not internalize the interest rate schedule). This occurs because the planner now internalizes that increasing debt issuance ($B_t \uparrow$) leads to higher interest rates ($r_{t+1} \uparrow$), which raises costs on all infra-marginal units of debt. Optimal current consumption will then be lower than in Case 1.

This behavior resembles a monopolist extracting rents by restricting quantity (debt issuance) to lower the interest rate paid on debt. The planner effectively exercises market power in the debt market, recognizing that additional debt issuance would drive up interest rates on the entire stock of outstanding debt. Overall, the planner has an incentive to *manipulate interest rates*: by reducing borrowing compared to Case 1, it can reduce the interest rate and hence extract rents from the foreigners.

Importantly, the strength of the rent extraction motive depends on the elasticity of RoW demand. If A_t^* is more elastic, then Γ_t decreases, leading to more debt issuance. Intuitively, when foreign investors are more responsive to interest rate changes, the planner faces less upward pressure on interest rates when issuing additional debt. In the limiting case of perfectly elastic foreign demand ($\Gamma_t = 0$), the solutions for Cases 1 and 2 converge, as the planner has no incentive to manipulate interest rates to extract rents from the RoW.

2.4 Case 3: Unrestricted Domestic Demand with Interest Rate Internalization

In our third case, the planner cannot discriminate between domestic and foreign agents, and domestic agents can freely purchase government bonds. The bond market clearing condition becomes $B_t = A_t + A_t^*$, where A_t represents domestic bond holdings.

Given its inability to segment the market, the planner internalizes the fact that the domestic agent's Euler equation holds. Therefore the sequence of interest rates is determined by domestic agent's Euler equation and the aggregate resource constraint:

$$\begin{aligned} 1 + r_{t+1} &= \frac{1}{\beta}\frac{U'(C_t)}{U'(C_{t+1})} \\ C_t &= Y_t + A_t^*(r_{t+1}) - (1 + r_t)A_{t-1}^*(r_t), \end{aligned}$$

that gives:

$$1 + r_{t+1} = \frac{1}{\beta}\frac{U'\Big(Y_t + A_t^*(r_{t+1}) - (1 + r_t)A_{t-1}^*(r_t)\Big)}{U'\Big(Y_{t+1} + A_{t+1}^*(r_{t+2}) - (1 + r_{t+1})A_t^*(r_{t+1})\Big)}. \tag{4}$$

It is clear from equation (4) that the equilibrium sequence of interest rates does not depend on the sequence of debt $\{B_t\}$ selected by the planner. Therefore, the planner loses the ability to manipulate interest rates and is hence indifferent between any path of debt issuance.

Interpretation. This third scenario presents a critical shift in the planner's ability to influence market outcomes. The planner loses the ability to manipulate the real interest rate through debt

issuance, as domestic agents' optimization decisions now determine the interest rate. This is consistent with Ricardian equivalence: when domestic agents can freely participate in bond markets, the path of government debt does not affect the real interest rate.

The solution in this case coincides with Case 1, meaning that the planner issues more debt than would be optimal under Case 2. This happens because the domestic agent takes the interest rate as given and does not internalize that her borrowing/savings decisions raise interest rates. Consequently, interest rates are inefficiently high in equilibrium, leading foreign investors to purchase more debt than is socially optimal.

This case demonstrates an important externality in debt markets. When domestic agents can freely purchase government bonds, they fail to account for how their actions collectively influence interest rates. The planner, despite understanding this relationship, cannot correct this market failure because it cannot exclude domestic agents from the bond market. As a result, the economy experiences over-borrowing from abroad and a reduction in welfare compared to case 2.[4]

Full insurance vs Interest rate manipulation. Under full domestic insurance, if a planner cannot discriminate between domestic and foreign agents buying its debt, it loses the ability to affect interest rates. Importantly, this shows that the planner cannot achieve simultaneously both full domestic insurance and optimal interest rate manipulation.

2.5 Welfare Implications

Our analysis reveals that Case 2, where the planner restricts domestic bond purchases and internalizes interest rate effects, yields the highest welfare among the three regimes. This welfare advantage occurs because the planner effectively exercises market power to secure more favorable borrowing terms. By limiting debt issuance below the level that would equate marginal utilities across periods according to the standard Euler equation, the planner reduces interest payments on the entire stock of debt.

In contrast, Cases 1 and 3 lead to excessive borrowing from abroad. In Case 1, the planner fails to account for how additional debt increases interest payments on existing debt. In Case 3, even though the planner understands this relationship, it cannot implement the optimal policy because it cannot restrict domestic participation in the bond market. The resulting equilibrium features higher interest rates and larger foreign bond holdings than optimal.

These results highlight the importance of institutional design in sovereign debt management. The ability to restrict domestic participation in government bond markets, combined with the understanding of how debt issuance affects interest rates, can improve welfare outcomes. However, such restrictions may be difficult to implement or maintain in practice, particularly in economies with developed financial markets and capital mobility.

The planner can therefore adopt a second best strategy of limiting the degree of domestic insurance in the economy. In particular, in the presence of uninsurable idiosyncratic risk, government debt is an effective tool to provide liquidity to domestic agents and partially insure them against risk (Aiyagari and McGrattan, 1998, Angeletos et al., 2023). In this context, even if domestic agents can buy the debt, the planner gains back the ability to affect the interest rate by changing the level of government debt. Therefore, the planner may be willing to provide less insurance to the economy by reducing its borrowing. In this way it will be able to manipulate interest rates and reduce them to extract rents from abroad. This comes at the cost of only partially insuring

[4]Notice that in case 3, the planner's problem has an additional constraint than case 2, i.e., the domestic agent's Euler equation. The optimal allocation in case 2 is not feasible in case 3.

domestic agents. Overall, the targets of full domestic insurance and optimal monopoly rents extraction are incompatible, introducing a meaningful trade-off for the planner that I will study and characterize in the following sections.

3 Characterization of the trade-off in a two-period model

In the previous section, I have shown how a planner that cannot discriminate between domestic and foreign holders of its government debt, faces the fundamental trade-off between providing domestic liquidity and optimally manipulating interest rates to extract monopoly rents from foreign bond holders. In this section I characterize this trade-off in a stylized heterogeneous agents model à la Woodford (1990). I will first focus on a two-period version.

3.1 Model setup

I consider here a simplified version of Woodford (1990). In particular I study a two-period and two-agents endowment economy with no risk.

Consumers. The economy runs for two periods (t and $t+1$) and features two agents (A and B) each with measure 1/2. The two agents receive alternating endowments over time. In particular, in each period one agent is poor since she receives zero income, while the other agent is rich since she receives an income equal to $2y_t$. We denote time t consumption of the agent who is rich in period t as $c_{r,t}$ and time t consumption of the agent who is poor in period t as $c_{p,t}$. The endowment and consumption pattern can be represented in the following table:

		t	$t+1$
Agent (A)	*Endowment*	$2y_t$	0
	Consumption	$c_{r,t}$	$c_{p,t+1}$
Agent (B)	*Endowment*	0	$2y_{t+1}$
	Consumption	$c_{p,t}$	$c_{r,t+1}$

Both agents can purchase domestic bonds $a_{i,t}$ but face a borrowing constraint $a_{i,t} \geq 0$, meaning they cannot borrow. Importantly, agents cannot issue and trade assets between them. They can only save in government risk-free debt. To avoid that the poor agent in t (agent B) starts with zero income and assets in t, we assume that she holds some initial government bonds $2A_{t-1}$, assumed to be small enough. These bonds pay an exogenous interest rate r_t, hence the initial financial income for the poor agent is $a_{p,-1} = (1+r_t)2A_{t-1}$. I impose the following assumption.

Assumption 1. *Assume that:*

$$(1+r_t)A_{t-1} < \frac{y_{t+1}}{\beta(1+r^{FI}_{t+1})},$$

where r^{FI}_{t+1} is determined implicitly by:

$$1+r^{FI}_{t+1} = \frac{1}{\beta}\frac{y_{t+1} - (1+r^{FI}_{t+1})A^*_t(r^{FI}_{t+1})}{y_t + A^*_t(r^{FI}_{t+1}) - (1+r_t)A^*_{t-1}}.$$

In other words, the initial level of assets of the poor agent has to be sufficiently smaller than her endowment in $t+1$.

This assumption allows partial insurance to be a possible equilibrium outcome by requiring that the poor agent gross income in $t+1$ (discounted using the full insurance equilibrium interest rate r_{t+1}^{FI}) is larger than its initial financial income. This makes the Euler equation of the poor agent be an inequality for sufficiently low level of government debt.

Finally, I assume that agents' utility function $u(\cdot)$ is logarithmic – $u(c_{i,t}) = \log(c_{i,t})$ for $i \in \{r, p\}$ – and future utility is discounted with discount factor $\beta \in (0, 1)$.

Rest of the World. As in Section 2, I assume that rest of the world (RoW) can save in domestic assets. In period t the RoW demands $A_t^*(r_{t+1})$ risk-free domestic assets, where r_{t+1} is the risk-free real interest rate between t and $t+1$. I assume that the demand for the domestic asset is differentiable and increasing in the real interest rate, hence $\frac{\partial A_t^*}{\partial r_{t+1}} > 0$.

Ramsey planner. A government issues risk-free debt B_t in period t paying the real interest rate r_{t+1} and in both periods $s = t, t+1$ imposes taxes proportional to endowment (τ_s is the tax rate).[5] In t the government has an initial stock of debt B_{t-1} and has to pay an exogenous interest rate r_t. The government budget constraints in periods t and $t+1$ are:

$$\begin{aligned} B_t &= (1 + r_t)B_{t-1} - \tau_t y_t, \\ 0 &= (1 + r_{t+1})B_t - \tau_{t+1} y_{t+1}. \end{aligned} \tag{5}$$

Notice that in the first period, since r_t and B_{t-1} are exogenous any increase in B_t will be used to reduce τ_t. In the second period instead taxes are raised to repay the existing stock of debt. Overall, the government has only one degree of freedom in its policy choices: issuing B_t determines τ_t and τ_{t+1}.

As described later more in detail, the government chooses B_t optimally as a Ramsey planner maximizing utilitarian welfare. Importantly, the government maintains full commitment to repay debt but cannot discriminate between domestic and foreign debt holders when selling bonds.

Similarities with Bewley models. Despite being stylized, this setup captures some key aspects of Bewley economies. In particular, a fraction of agents will be constrained and featuring high MPCs. Importantly for our analysis, as in Bewley economies, the aggregate asset demand from consumers is increasing in the interest rate. Hence, an increase in the issuance of debt will increase the interest rate.

A difference with respect to a general Bewley model lies in the fact that our two agents have a degenerate Markov transition kernel between periods since poor agents become rich with probability one and viceversa. This means that agents save to smooth consumption intertemporally but the poor agent is prevented from borrowing due to borrowing constraints. However, differently from Bewley models, agents do not have an additional precautionary savings motive. Consistently with the absence of this additional motive, the closed economy Ramsey steady state of the Woodford model (see the infinite horizon version in Section 4) features the gross interest rate to equal the inverse of the discount factor β^{-1}.[6] See also Auclert, Cai, Rognlie, and Straub (WP) for a discussion about the Woodford model relative to general Bewley-Aiyagari frameworks.

[5] In this endowment economy this is equivalent to assume lump-sum taxation, not uniform between agents.

[6] In general, the steady-state assets distribution is not defined for this level of the interest rate.

3.2 Consumer Problem

Consumers face the usual intertemporal maximization problem. Each agent $j \in \{A, B\}$ maximizes lifetime utility under budget and borrowing constraints:

$$\begin{aligned} \max_{c_{j,t}, c_{j,t+1}, a_{j,t}} \quad & \log(c_{j,t}) + \beta \log(c_{j,t+1}) \\ \text{s.t.} \quad & c_{j,t} + a_{j,t} = y_{j,t}(1-\tau_t) + (1+r_t)a_{j,t-1}, \\ & c_{j,t+1} = y_{j,t+1}(1-\tau_{t+1}) + (1+r_{t+1})a_{j,t}, \\ & a_{j,t} \geq 0, \quad a_{j,t-1} \text{ given.} \end{aligned}$$

Here I solve the optimization problems for two cases: (i) partial insurance (assuming that poor agents are constrained); (ii) full insurance (both agents are not constrained). I will then show that whether the economy is in (i) or (ii) depends on the level of debt issuance only.

3.2.1 Partial insurance

Poor agent. Under partial insurance, the poor agent at time t (agent B) faces a binding borrowing constraint. Her marginal utility of current consumption exceeds the discounted expected marginal utility of future consumption, adjusted by the interest rate:

$$u'(c_{p,t}) > \beta(1+r_{t+1})u'(c_{r,t+1}).$$

Therefore, her optimal consumption and savings are:

$$a_{p,t} = 0, \tag{6}$$

$$c_{p,t} = (1+r_t)2A_{t-1}, \tag{7}$$

$$c_{r,t+1} = 2y_{t+1}(1-\tau_{t+1}). \tag{8}$$

Rich agent. For the rich agent at time t (agent A) the standard Euler equation holds:

$$u'(c_{r,t}) = \beta(1+r_{t+1})u'(c_{p,t+1}).$$

Using the intertemporal budget constraint

$$c_{r,t} + \frac{c_{p,t+1}}{1+r_{t+1}} = 2y_t(1-\tau_t),$$

I solve for rich agents optimal consumption and savings:

$$a_{r,t} = \frac{\beta}{1+\beta} 2y_t(1-\tau_t), \tag{9}$$

$$c_{r,t} = \frac{1}{1+\beta} 2y_t(1-\tau_t), \tag{10}$$

$$c_{p,t+1} = (1+r_{t+1})a_{r,t} = (1+r_{t+1})\frac{\beta}{1+\beta} 2y_t(1-\tau_t). \tag{11}$$

As standard with log-utility, the rich agent consumes a fraction of his lifetime income $2y_t(1-\tau_t)$ and saves the rest. I define aggregate domestic assets demand as:

$$A_t := \frac{1}{2}a_{p,t} + \frac{1}{2}a_{r,t},$$

that gives:

$$A_t = \frac{\beta}{1+\beta} y_t (1-\tau_t). \tag{12}$$

Notice that the asset demand does not depend on the interest rate r_{t+1}. This is a consequence of log-utility and zero future endowment for the rich agent. In this context substitution and income effects from interest rate changes cancel out. Overall, domestic asset demand only depends on the current period tax rate.[7]

Aggregate consumption at time t – defined by $C_t := \frac{1}{2}c_{p,t} + \frac{1}{2}c_{r,t}$ – is instead given by:

$$C_t = \frac{1}{1+\beta} 2y_t(1-\tau_t) + (1+r_t)A_{t-1}. \tag{13}$$

3.2.2 Full insurance

Under full insurance both agents are on their Euler equation. Therefore we have:

$$u'(c_{r,t}) = \beta(1+r_{t+1})u'(c_{p,t+1}),$$
$$u'(c_{p,t}) = \beta(1+r_{t+1})u'(c_{r,t+1}).$$

As usual, this implies that the consumption ratios between agents are constant over time:

$$\frac{c_{p,t}}{c_{r,t}} = \frac{c_{r,t+1}}{c_{p,t+1}} = \text{constant}.$$

While the optimal consumption and savings for the time t rich agent is as before – equations (10, 11, 9) –, poor agents are also optimizing over two periods now. Therefore their policy functions are given by:

$$a_{p,t} = \frac{1}{1+\beta}\left[\beta(1+r_t)2A_{t-1} - \frac{2y_{t+1}(1-\tau_{t+1})}{1+r_{t+1}}\right],$$
$$c_{p,t} = \frac{1}{1+\beta}\left[(1+r_t)2A_{t-1} + \frac{2y_{t+1}(1-\tau_{t+1})}{1+r_{t+1}}\right],$$
$$c_{r,t+1} = \frac{\beta(1+r_{t+1})}{1+\beta}\left[(1+r_t)2A_{t-1} + \frac{2y_{t+1}(1-\tau_{t+1})}{1+r_{t+1}}\right].$$

Hence aggregate asset demand is:

$$A_t = \frac{1}{1+\beta}\left(\beta y_t - \frac{y_{t+1}}{1+r_{t+1}}\right) + \frac{\beta}{1+\beta}[(1+r_t)A_{t-1} - \tau_t y_t] + \frac{1}{1+\beta}\frac{\tau_{t+1}y_{t+1}}{1+r_{t+1}}. \tag{14}$$

Notice that under full insurance the domestic aggregate asset demands depends on the interest rate and both current and future tax rates. Aggregate consumption at time t is given by:

$$C_t = \frac{1}{1+\beta}\left[(1+r_t)A_{t-1} + y_t(1-\tau_t) + \frac{y_{t+1}(1-\tau_{t+1})}{1+r_{t+1}}\right]. \tag{15}$$

[7] While convenient for analytical purposes, this simplification does not change our results.

3.2.3 Aggregate domestic assets demand

The economy is in partial insurance as long as the poor agent's Euler equation does not hold with equality:

$$c_{r,t+1} \geq \beta(1 + r_{t+1})c_{p,t}$$

Substituting for (7), (8) and for the government budget constraint (5) we get that the inequality holds if and only if:

$$B_t \leq \hat{B}_t(r_{t+1}) := \frac{y_{t+1}}{1 + r_{t+1}} - \beta(1 + r_t)A_{t-1}.$$

Hence $\hat{B}_t(r_{t+1})$ denotes the threshold level of debt at which poor agents become fully insured. Below that level of debt, the economy is instead partially insured. Notice that $\hat{B}_t$ depends on the interest rate r_{t+1} that is determined in equilibrium.

3.3 Competitive equilibrium

The competitive equilibrium is an allocation $\{c_{i,t}, a_{i,t}\}$, an interest rate r_{t+1} and tax rates τ_t, τ_{t+1} such that – given B_t –:

- Domestic agents solve their optimization problem and resulting aggregate consumptions and assets are given as in (13), (12) or (15), (14).
- Government budget constraints (5) hold.
- Asset market clearing requires:
$$B_t = A_t + A_t^*.$$
- Goods market clearing requires:
$$C_t = y_t + A_t^* - (1 + r_t)A_{t-1}^*,$$
$$C_{t+1} = y_{t+1} - (1 + r_{t+1})A_t^*.$$

The competitive equilibrium is defined for a given level of debt. In fact, given B_t taxes are determined by the government budget constraint. In turn assets demand is also determined from the agents' problem. Finally, the interest rate clears the asset market. Overall, in the two-period economy the level of debt parametrizes the competitive equilibrium. The Ramsey problem will then amount to the one-dimensional choice of the level of debt that maximizes utilitarian welfare.

3.4 Equilibrium Interest Rate

The relationship between interest rates and debt levels is central to my analysis. I derive now the parametric form of the interest rate schedule as a function of debt: $r_{t+1} = \mathcal{R}_t(B_t)$.

Domestic asset demand. Using the government budget constraints (5) I can express aggregate domestic assets as:

$$A_t(r_{t+1}, B_t) = \begin{cases} \frac{\beta}{1+\beta}[y_t - (1 + r_t)B_{t-1}] + \frac{\beta}{1+\beta}B_t & \text{if } B_t \leq \hat{B}_t, \\ \\ \frac{1}{1+\beta}\left(\beta[y_t - (1 + r_t)A_{t-1}^*] - \frac{y_{t+1}}{1+r_{t+1}}\right) + B_t & \text{if } B_t > \hat{B}_t. \end{cases} \tag{16}$$

For a given interest rate $A_t(r_{t+1}, B_t)$ is continuous and increasing in the level of debt, but less than one-to-one for $B_t \leq \hat{B}_t$ and one-to-one for $B_t > \hat{B}_t$. The fact that A_t is increasing in B_t is immediate to verify. I show that $A_t(r_{t+1}, B_t)$ is continuous in B_t in Appendix A.2.

Asset market equilibrium. Imposing the assets market clearing $B_t = A_t(r_{t+1}, B_t) + A_t^*(r_{t+1})$ determines the interest rate. When the economy is at full insurance, for a sufficiently high level of debt, the equilibrium interest rate does not depend on the level of debt, consistent with Ricardian equivalence. In particular, the full insurance interest rate is determined implicitly:

$$\frac{1}{1+\beta}\left(\beta y_t - \frac{y_{t+1}}{1+r_{t+1}^{FI}}\right) - \frac{\beta}{1+\beta}A_{t-1}^*(1+r_t) + A_t^*(r_{t+1}^{FI}) = 0$$

that can be equivalently expressed as:

$$1 + r_{t+1}^{FI} = \frac{1}{\beta}\frac{y_{t+1} - (1+r_{t+1}^{FI})A_t^*(r_{t+1}^{FI})}{y_t + A_t^*(r_{t+1}^{FI}) - (1+r_t)A_{t-1}^*} = \frac{1}{\beta}\frac{C_{t+1}^{FI}}{C_t^{FI}}, \tag{17}$$

that relates the interest rate to aggregate consumption growth. In fact, under full insurance it is possible to aggregate consumption from the Euler equations of both agents and hence express the interest rate as a function of aggregate consumption growth. Notice also that the presence of foreign demand reduces the equilibrium interest rate (compared to the closed economy counterpart).

When the economy is at partial insurance, the interest rate is determined by the following equilibrium condition:

$$A_t^*(r_{t+1}) = -\frac{\beta}{1+\beta}y_t + \frac{\beta}{1+\beta}(1+r_t)B_{t-1} + \frac{1}{1+\beta}B_t.$$

From the equation above, the interest rate is an increasing function of B_t with elasticity:

$$\Gamma_t := \frac{\partial r_{t+1}}{\partial \log B_t}\frac{1}{1+r_{t+1}} = \frac{1}{(1+\beta)(1+r_{t+1})}\frac{1}{s_t^* \cdot \xi_t^*}, \tag{18}$$

where s_t^* is the share of debt held by RoW and ξ_t^* is the elasticity of RoW assets demand:

$$s_t^* := \frac{A_t^*}{B_t}, \quad \xi_t^* := \frac{\partial \log A_t^*}{\partial r_{t+1}}.$$

Consistent with our discussion in Section 3.2.3, the economy is at partial insurance as long as:

$$B_t \leq \hat{B}_t(r_{t+1}^{FI}) = \frac{y_{t+1}}{1+r_{t+1}^{FI}} - \beta A_{t-1}(1+r_t). \tag{19}$$

Overall, the interest rate schedule is increasing when the economy is at partial insurance – for $B_t \leq \hat{B}_t(r_{t+1}^{FI})$ – and constant at r_{t+1}^{FI} thereafter.

3.5 The Ramsey Problem

The planner solves a Ramsey problem by choosing debt B_t, tax rates τ_t and τ_{t+1}, and the interest rate r_{t+1} to maximize utilitarian welfare. Future utility streams are discounted with private domestic

agents' discount factor β (more general discounting will be discussed in Section 4). The planner's objective is therefore:

$$W_t = \sum_{s=0}^{1} \beta^s [u(c_{r,t+s}) + u(c_{p,t+s})].$$

The planner operates under a set of constraints, including domestic agents' policy functions, the economy resource constraints, and equilibrium conditions. From the government budget constraints (5), it is immediate to notice that the choice of debt determines τ_t, τ_{t+1}: the planner has hence only one degree of freedom. In the case the planner selects partial insurance equilibria (i.e., when debt is below its full insurance level $B_t \leq \hat{B}_t$), its optimization problem can be simplified to the following:[8]

$$\begin{aligned}
\max_{B_t} \quad & \sum_{s=0}^{1} \beta^s [u(c_{r,t+s}) + u(c_{p,t+s})] \\
\text{s.t.} \quad & \frac{1}{2} c_{r,t} + \frac{1}{2} c_{p,t} = y_t + A_t^*(r_{t+1}) - (1 + r_t) A_{t-1}^*, \\
& \frac{1}{2} c_{r,t+1} + \frac{1}{2} c_{p,t+1} = y_{t+1} - (1 + r_{t+1}) A_t^*(r_{t+1}), \\
& c_{p,t+1} = (1 + r_{t+1}) 2 [B_t - A_t^*(r_{t+1})], \\
& r_{t+1} = \mathcal{R}_t(B_t).
\end{aligned}$$

The first two constraints are the aggregate resource constraints of the economy. The third constraints is future consumption of poor agents (i.e., the rich agent in period t) that consumes its assets from the previous period:

$$c_{p,t+1} = (1 + r_{t+1}) 2 A_t,$$

where $A_t = B_t - A_t^*$. Finally, the planner takes as a constraint the equilibrium interest rate schedule $r_{t+1} = \mathcal{R}_t(B_t)$, as discussed in Section 3.4.

3.6 Solution and Interpretation

Solving the planner's problem I can express the change in welfare relative to debt as:[9]

$$\frac{dW_t}{dB_t} = \underbrace{\beta [u'(c_{p,t+1}) - u'(c_{r,t+1})] \cdot (1 + r_{t+1})(1 + \Gamma_t)}_{\text{Redistribution/insurance term}} - \underbrace{\beta u'(c_{p,t+1}) \cdot (1 + r_{t+1}) s_t^* \Gamma_t}_{\text{Interest rate manipulation term}}. \tag{20}$$

At the optimum $\frac{dW_t}{dB_t} = 0$. The change in welfare can be decomposed in the contribution of two terms: a redistribution/insurance and a rent extraction term.

- *Redistribution/insurance* term. As the planner increases debt, it indirectly redistributes income from the time $t+1$ rich to the time $t+1$ poor agent. In fact, the time $t+1$ rich agent has to pay higher taxes to finance the higher stock of debt, while the time $t+1$ poor agent has more assets from the previous period. This redistribution channel includes both the direct effect of increasing debt and its indirect effect on higher interest rates, captured by the elasticity Γ_t. In the following section, I will decompose this term and study its sign.

[8] If the planner selects also full insurance equilibria it additionally takes the optimal choice for $c_{p,t}$ as a constraint. See Appendix A.1 for details on the general Ramsey problem.

[9] See Appendix A.2 for the solution to the planner's problem.

- *Interest rate manipulation* term. As the planner increases debt, the aggregate resource constraint in period $t+1$ tightens since the economy has to pay higher interests to foreigners. The additional cost is described by the term $(1+r_{t+1})s_t^*\Gamma_t$ and is hence proportional to the share of debt held by the RoW. In welfare terms the cost is evaluated by the discounted marginal utility of time $t+1$ poor agents (time t rich agent) $\beta u'(c_{p,t+1})$, since she is the one holding and hence pricing government debt between t and $t+1$.

3.6.1 Intuitive decomposition

It is possible to decompose the *redistribution/insurance* term into a pure insurance and pure redistribution components.[10] Moreover, using the time t rich agent's Euler equation I can re-express also the *interest rate manipulation* term as a wedge in their Euler equation. Overall, the change in welfare can be expressed as:

$$\begin{aligned}\frac{dW_t}{dB_t} = & \underbrace{[u'(c_{p,t}) - \beta(1+r_{t+1})u'(c_{r,t+1})]}_{\text{Partial insurance term, >0}}(1+\Gamma_t) \quad + \quad \underbrace{[u'(c_{r,t}) - u'(c_{p,t})]}_{\text{Residual inequality term, }\gtrless 0} \quad (1+\Gamma_t) \\ & + \underbrace{u'(c_{r,t}) - \beta(1+r_{t+1})u'(c_{p,t+1})(1+s_t^*\Gamma_t)}_{\text{Interest rate manipulation term, <0}}.\end{aligned} \tag{21}$$

This expression reveals three key forces that determine the optimal debt level. First, the partial insurance term captures the welfare benefit of providing liquidity to credit-constrained agents. The term $[u'(c_{p,t}) - \beta(1+r_{t+1})u'(c_{r,t+1})]$ represents the wedge in the Euler equation of the poor agent. Higher debt provides liquidity to poor agents, helping to close this wedge and improve welfare. This term pushes toward higher debt issuance.

Second, the residual inequality term captures welfare losses from consumption inequality even under full insurance. In fact, even under full insurance there may be consumption inequality across agents. If initial assets held by time t poor agents A_{t-1} is sufficiently small, this term is non-positive as $c_{r,t} \geq c_{p,t}$ for debt levels close to the full insurance one (i.e., close to $\hat{B}_t$).[11] In this case the residual inequality term pushes the planner to reduce debt from the full insurance level. In the opposite case where this term is positive the planner may be willing to increase debt above $\hat{B}_t$.

Third, the interest rate manipulation term reflects the benefit of reducing debt to lower interest rates. This term recalls the representative agent condition (3) from the first section of our analysis, where the planner internalizes how debt affects interest rates. The planner has an incentive to reduce debt, thereby reducing the interest rate r_{t+1} and the cost of on infra-marginal units of debt. This term resembles the representative agent economy optimality condition (3), with two differences: (i) the relevant marginal utility is that of time t rich agent since she is the one pricing the asset in the partial insurance case; (ii) the effect of changes in the interest rate is proportional to the share s_t^* of foreign held debt.

3.6.2 Role of openness and RoW elasticity

Closed economy. We compare the Ramsey plan in the open economy to the closed economy counterpart. In a closed economy, $s_t^* = 0$, hence equation (21) becomes:

$$\frac{dW_t}{dB_t} = [u'(c_{p,t}) - \beta(1+r_{t+1})u'(c_{r,t+1})](1+\Gamma_t) + [u'(c_{r,t}) - u'(c_{p,t})](1+\Gamma_t).$$

[10] See Appendix A.3.

[11] Details in Appendix A.3.

Moreover, it is possible to show that only full insurance levels of debt are supported in equilibrium.[12] See Appendix A.4 for details. In other words, a closed economy Ramsey planner will optimally set:

$$B_t \geq \hat{B}_t(r_{t+1}^{FI}), \quad 1 + r_{t+1}^{FI} = \frac{1}{\beta}\frac{y_{t+1}}{y_t}.$$

If at $B_t = \hat{B}_t$, $c_{r,t} < c_{p,t}$ then the planner will then set an optimal level of debt above $\hat{B}_t$ to obtain full redistribution between the two agents:[13]

$$u'(c_{r,t}) = u'(c_{p,t}).$$

Otherwise if at $B_t = \hat{B}_t$, $c_{r,t} \geq c_{p,t}$, then the planner will optimally set $B_t = \hat{B}_t$.

Foreign demand elasticity. In the open economy I have shown that the elasticity of the interest rate to debt is equal to:

$$\Gamma_t = \frac{1}{(1+\beta)(1+r_{t+1})}\frac{1}{s_t^*\xi_t^*},$$

where ξ_t^* is the elasticity of RoW demand for domestic debt. It is clear that Γ_t is a decreasing function of ξ_t^*. The more elastic foreign demand is, the smaller the interest rate manipulation term is and the larger the weight the planner will allocate to the insurance/redistribution motives. Intuitively, with more elastic demand the planner is able to manipulate interest rates without reducing debt issuance too much compared to the full insurance level.

In the limit case of a perfectly elastic RoW demand, $\Gamma_t = 0$ and the interest rate manipulation term disappears. The planner will then optimally provide liquidity to bring the economy at full insurance (and indirectly redistribute consumption between agents).[14]

3.6.3 Summary of results

My analysis yields three important findings:

1. The optimal debt level is below what would be required for full insurance and optimal domestic liquidity provision. This indicates that the planner optimally tradeoffs some domestic liquidity provision to extract monopoly rents from foreign investors.

2. More inelastic foreign demand intensifies this trade-off. When foreigners' demand for domestic bonds is less responsive to interest rates, the planner has a stronger incentive to restrict debt issuance to manipulate rates.

3. In a closed economy ($s_t^* = 0$) or with perfectly elastic foreign demand ($\Gamma_t = 0$), the interest rate manipulation term disappears and the planner focuses solely on domestic liquidity provision and redistribution.

[12] This depends on the fact that under partial insurance, time t rich agent's asset demand does not depend on the interest rate. A more general extension of the model where rich agents receive some endowment also in $t+1$ will allow for closed-economy equilibria under partial insurance.

[13] Since by increasing debt it will increase taxes in $t+1$ and hence redistribute income from the time t poor agent to the rich one.

[14] In the other limit case of perfectly inelastic RoW demand it is also possible to show – as for the closed economy case – that only levels of debt providing full insurance are possible equilibria. This feature depends on the fact that domestic asset demand under partial insurance does not depend on the interest rate. See Appendix A.4 for details.

This analysis demonstrates that open economies face a fundamental trade-off between providing domestic liquidity and extracting monopoly rents from foreign investors. The optimal policy balances these competing objectives, with the balance depending critically on the elasticity of foreign demand.

4 Characterization of the trade-off in an infinite horizon model

To extend my analysis beyond the two-period framework, I now consider an infinite horizon model with alternating incomes, following Woodford (1990). This model preserves the key features of the two-period case while allowing to characterize long-run equilibrium properties and derive steady-state implications of the trade-off between liquidity provision and interest rate manipulation.

4.1 Model Setup

The economy consists of two agents (each with measure 1/2) who alternate between high and low income states. Time is discrete.

Consumers. In every period t one agent is rich and receives an endowment of $2y_t$ while the other agent is poor and receives nothing. In the following period the endowment switches across the two agents and keeps alternating in every period. The endowment pattern can be summarized in the following table.

	t	$t+1$	$t+2$	$t+3$	$t+4$
Agent (A)	$2y_t$	0	$2y_{t+2}$	0	...
Agent (B)	0	$2y_{t+1}$	0	$2y_{t+3}$	...

I denote by $c_{r,t}$ consumption of the agent who receives the high income in period t (the rich agent) and by $c_{p,t}$ consumption of the agent who receives zero income in period t (the poor agent).

Each agent can purchase domestic bonds $a_{i,t}$ paying a risk-free real interest rate $1 + r_{t+1}$, but faces the borrowing constraints $a_{i,t} \geq 0$. Finally, I assume that agents' have logarithmic utility and discount the future with a discount factor $\beta \in (0, 1)$.

Rest of the World. As in Section 3.1, I assume that the rest of the world (RoW) can save in domestic debt and denote its demand by $A_t^*(r_{t+1})$.

Ramsey planner. A government issues risk-free debt B_t paying the real interest rate r_{t+1} (between t and $t+1$) and chooses period t tax rate τ_t (proportional to agents' endowment). The government budget constraint in period t is:[15]

$$B_t = (1 + r_t)B_{t-1} - \tau_t y_t. \tag{22}$$

The government chooses the sequence of debt, taxes, and interest rates $\{B_t, \tau_t, r_{t+1}\}$ optimally as a Ramsey planner maximizing utilitarian welfare for a general discount factor $\delta \in [\beta, 1]$. The government maintains full commitment to repay debt but cannot discriminate between domestic and foreign debt holders when selling bonds.

[15] There is an initial condition for debt B_{-1} and interest rate r_0.

4.1.1 Consumer problem

Consider the partial insurance scenario where (for every period t) the time t poor agent is liquidity constrained. Then the poor agent only consumes her assets from the previous period. Instead, the time t rich agent solves a two-period problem, anticipating that she will be constrained between $t+1$ and $t+2$. Overall, agents' optimal policy functions are no different than our two-period model of Section 3.2.

Poor agent. Under partial insurance, the time t poor agent's Euler equation does not hold ($u'(c_{p,t}) > \beta(1+r_{t+1})u'(c_{r,t+1})$) and her optimal consumption and savings are:

$$\begin{aligned} a_{p,t} &= 0, \\ c_{p,t} &= (1+r_t)2A_{t-1}. \end{aligned}$$

Rich agent. The standard Euler equation and intertemporal budget constraint hold:

$$\begin{aligned} u'(c_{r,t}) &= \beta(1+r_{t+1})u'(c_{p,t+1}), \\ c_{r,t} + \frac{c_{p,t+1}}{1+r_{t+1}} &= 2y_t(1-\tau_t). \end{aligned}$$

Rich agents optimal consumption and savings are:

$$\begin{aligned} a_{r,t} &= \frac{\beta}{1+\beta}2y_t(1-\tau_t), \\ c_{r,t} &= \frac{1}{1+\beta}2y_t(1-\tau_t). \end{aligned}$$

4.1.2 Aggregation and competitive equilbrium

I define aggregate domestic assets demand as $A_t := \frac{1}{2}a_{p,t} + \frac{1}{2}a_{r,t}$ and aggregate consumption as $C_t := \frac{1}{2}c_{p,t} + \frac{1}{2}c_{r,t}$. In the partial insurance equilibrium they are equal to:

$$A_t = \frac{\beta}{1+\beta}y_t(1-\tau_t), \tag{23}$$

$$C_t = \frac{1}{1+\beta}2y_t(1-\tau_t) + (1+r_t)A_{t-1}. \tag{24}$$

Notice that aggregate assets demand in the partial insurance case is a function of the tax rate τ_t only.

Competitive equilibrium. For a given sequence of debt and tax rates $\{B_t, \tau_t\}_{t=0}^{\infty}$, the competitive equilibrium is an allocation $\{c_{i,t}, a_{i,t}, B_t\}_{t=0}^{\infty}$ and a sequence of interest rates $\{r_{t+1}\}_{t=0}^{\infty}$ such that:

- Domestic agents solve their optimization problem.
- Government budget constraints (22) hold.
- Assets market clears: $B_t = A_t + A_t^*$.
- Goods market clears: $C_t = y_t + A_t^* - (1+r_t)A_{t-1}^*$.

4.2 The General Ramsey Problem

The planner's problem in the infinite horizon setting involves choosing the sequence of tax rates $\{\tau_t\}_{t=0}^{\infty}$ and interest rates $\{r_{t+1}\}_{t=0}^{\infty}$ to maximize the discounted sum of agents' utilities. Importantly, the planner may use a general social discount factor $\delta \in [\beta, 1]$, which may differ from the private discount factor β. This allows for a more flexible weighting of current versus future generations in the social welfare function. As argued in Auclert, Cai, Rognlie, and Straub (WP), a discount factor of $\delta = 1$ is equivalent to having the planner maximizing welfare in the steady state (in other words, the planner would only care about long-run outcomes). Overall, as δ increases the planner focuses more on long-run welfare.

Notice that the sequence of debt is determined from the government budget constraints (and policy functions), given a sequence of taxes and interest rates (and initial conditions). Hence, the planner only faces a sequence of implementability constraints that combine the government budget constraints with market clearing conditions and aggregate agents assets policy functions. When the sequence of tax and interest rates is such that the economy is partially insured in every period, aggregate domestic demand is given by (23), hence only depends on τ. Therefore, in the partial insurance case, the time t implementability constraint reads:

$$A_t(\tau_t) + A_t^*(r_{t+1}) = (1 + r_t)[A_{t-1}(\tau_{t-1}) + A_{t-1}^*(r_t)] - \tau_t y_t. \tag{25}$$

I characterize the Ramsey planner in the region of the space of the tax and interest rate sequences that support partial insurance equilibria in every period. In this region, the relevant implementability constraint is given by (25). Hence, the Ramsey problem can be formulated as:

$$\begin{aligned}
\max_{\{\tau_t, r_{t+1}\}_{t=0}^{\infty}} \quad & \sum_{t=0}^{\infty} \delta^t [u(c_{r,t}) + u(c_{p,t})] \\
\text{s.t.} \quad & c_{p,t} = (1 + r_t) 2 A_{t-1}(\tau_{t-1}), \\
& c_{r,t} = \frac{1}{1+\beta} 2(1 - \tau_t) y_t, \\
& A_t(\tau_t) = \frac{\beta}{1+\beta} y_t (1 - \tau_t), \\
& A_t(\tau_t) + A_t^*(r_{t+1}) = (1 + r_t)[A_{t-1}(\tau_{t-1}) + A_{t-1}^*(r_t)] - \tau_t y_t.
\end{aligned}$$

The first two constraints simply are the consumption policy functions of poor and rich agents respectively, the third constraint is aggregate domestic assets demand, while the last constraint is the implementability condition.

4.3 First-Order Conditions

The solution to the Ramsey problem yields two first-order conditions that characterize the optimal policy. Let λ_t denote the Lagrange multiplier on the implementability constraint at time t, and $\xi_t^* = \frac{\partial \log A_t^*}{\partial r_{t+1}}$ represent the elasticity of foreign demand with respect to the interest rate. The first-

order conditions with respect to τ_t and r_{t+1} are:[16]

$$[\tau_t]: \; 2(1+\delta)u'(c_{r,t}) = \lambda_t + \beta\delta\lambda_{t+1}(1+r_{t+1}), \tag{26}$$

$$[r_{t+1}]: \; \underbrace{\left[u'(c_{p,t+1}) - \frac{\lambda_{t+1}}{2}\right]2\delta A_t}_{\text{Insurance/redistribution}} + \underbrace{A_t^*\Big[\lambda_t - \delta\lambda_{t+1}(1+r_{t+1})\Big]\xi_t^* - \delta A_t^*\lambda_{t+1}}_{\text{Net liquidity benefits from RoW}} = 0. \tag{27}$$

The first condition (26) describes the trade-off associated with changing the tax rate. An increase in τ_t generates a marginal welfare loss (the left-hand side) since rich agents have lower lifetime income and cut current and future consumption (since they are on the Euler equation). Higher τ_t also provides a marginal gain (the right-hand side) from relaxing the implementability constraint by increasing current tax revenue and reducing future debt obligations.

The second condition (27) captures the trade-off associated with changing the interest rate. An increase in r_{t+1} affects welfare through two channels.

1. *Insurance and redistribution* channel: higher interest rates benefit poor agents in $t+1$ by raising the interest income on their assets. This marginal benefit is described by $\delta 2u'(c_{p,t+1})A_t$. On the contrary, higher interest payments from the government tighten the future implementability condition. This marginal cost is described by $-\delta\lambda_{t+1}A_t$.

2. *Net liquidity benefits from RoW* channel: captures the net benefits from changes in interest payments to the RoW and from changes in net foreign assets positions. First, higher interest rates impose a cost from higher interest payments on infra-marginal units of net foreign debt (described by $-\delta\lambda_{t+1}A_t^*$). Moreover, an increase in r_{t+1} rises foreign demand (A_t^*): this induces a benefit from relaxing current implementability constraint (described by $\lambda_t\xi_t^*A_t^*$), and a cost from tightening the implementability constraint in $t+1$ (described by $-\delta\lambda_{t+1}(1+r_{t+1})\xi_t^*A_t^*$).

4.4 Ramsey Steady State

I characterize the long-run properties of the optimal policy. The notion that I adopt is that of Ramsey State State (RSS), i.e., the limit of the sequence of optimal tax and interest rates that constitute the Ramasey plan.[17]

Consider a converging sequence of output: $\{y_t\}_{t=0}^{\infty} \longrightarrow y = 1$. First, I assume that a Ramsey plan exists, hence it satisfies the first-order conditions (26, 27). Second, I assume that the Ramsey plan converges to a steady state: $\{\tau_t, r_{t+1}\}_{t=0}^{\infty} \longrightarrow (\tau, r)$, that is indeed the RSS. Third, I assume convergence in the Lagrange multipliers $\{\lambda_t\}_{t=0}^{\infty} \longrightarrow \lambda$.

To characterize the RSS, notice that assets and consumption policy functions converge. Using equation (26), λ is:

$$\lambda_t \longrightarrow \lambda = \frac{2(1+\delta)u'(c_r)}{1+\beta\delta(1+r)}.$$

Finally, combining λ with equation (27) in the RSS, I derive the following necessary condition, characterizing the RSS:[18]

$$\underbrace{[1-\beta(1+r)]\frac{\delta}{(1+\beta)(1+r)}(1-\tau)}_{\text{Insurance/redistribution}} \underbrace{-(1+\delta)A^*\Big(\delta - [1-\delta(1+r)]\xi^*\Big)}_{\text{Net liquidity benefits from RoW}} = 0. \tag{28}$$

[16] See Appendix B.1 for derivations.
[17] See Auclert, Cai, Rognlie, and Straub (WP) for the study of RSS in general Aiyagari economies.
[18] See Appendix B.2 for derivations.

This condition reveals the central trade-off faced by the planner in the RSS. The first term in equation is related to the *insurance/redistribution* motive described in the previous section. In fact, the degree of domestic insurance is captured by the term $1 - \beta(1+r)$. If $\beta(1+r) = 1$ the economy is fully insured and consumption of rich and poor agents is equalized:

$$c_r = c_p \iff \beta(1+r) = 1$$

as it can be verified by the Euler equations. On the contrary, when $\beta(1+r) < 1$, the poor agent is constrained and $c_r > c_p$ and the economy is only partially insured. In the following section, I will show that partial insurance is optimal in the RSS.

The second term captures net liquidity benefits from RoW in the steady-state, as described in the previous section. This term is typically larger in absolute value for higher values of δ, since the planner weights welfare losses from steady-state interest payments to foreigners more than the impact benefits from higher debt sold to them.

Finally, together with equation (28) the RSS is characterized by the limit of the sequence of implementability constraints, that reads:

$$r(A(\tau) + A^*(r)) = \tau. \tag{29}$$

4.4.1 The Key Trade-off

Studying the necessary condition (28), I can show that a Ramsey planner will optimally choose partial insurance when the economy is open. I also explore how the solution depends on the discount factor δ. I summarize the results in the following proposition.

Proposition 1. *Consider the RSS described by conditions (28) and (29).*
(A) In an open economy ($A^* > 0$), for every $\delta \in [\beta, 1]$, the optimal RSS interest rate r is such that the economy is partially insured, hence:

$$\beta(1+r) < 1.$$

(B) In a closed economy ($A^* = 0$), for every $\delta \in [\beta, 1]$, the optimal RSS interest rate r is such that the economy is fully insured, hence:

$$\beta(1+r) = 1.$$

(C) In an open economy, the RSS interest rate r is decreasing in δ.[19]

Proof. See Appendix B.2. □

The first part of the proposition establishes that in an open economy the planner will only partially insure the domestic economy. Consider, for instance, the case of the $\delta = 1$ (meaning that the planner only cares about long-run outcomes). In this case equation (28) becomes:

$$\begin{aligned}
&[1 - \beta(1+r)]\frac{1}{(1+\beta)(1+r)}(1-\tau) - 2A^*(1 + r \cdot \xi^*) = 0. \\
\implies &[1 - \beta(1+r)](1-\tau) = 2A^*(1 + r \cdot \xi^*)(1+\beta)(1+r) \\
\implies &\beta(1+r) < 1.
\end{aligned}$$

[19] Under the additional assumptions that: (i) $\beta \geq \sqrt{2}/2 - 1/2$, (ii) A^* is twice-differentiable; (iii) $\partial^2 A^*/\partial r^2$ is locally bounded. See Assumption A.3 for details. Assumption (i) is requires $\sqrt{5}/2 - 1/2 \approx 0.62$ and is needed to impose bounds on the marginal changes in the net liquidity benefits term. (ii) and (iii) impose requirements on the curvature of the RoW demand around the solution of equation (28), but allows flexibility on whether A^* is concave or convex.

Intuitively, when $\delta = 1$ net benefits from abroad are always negative, because the planner focuses only on long-run outcomes. In the long-run, higher debt held by the RoW requires larger interest payments to foreigners. On the contrary, the planner is not incorporating the short-run benefits from higher debt sold to the RoW. Overall, since higher interest rates impose a long-run cost, the planner will optimally reduce the interest rate below the full insurance level in such a way to reduce the amount of debt bought by foreigners.

Turning to the case of $\delta < 1$, the planner now discounts future costs coming from interest payments more, while it cares more about the benefits coming from higher resources in the short-run (since higher interest rates increase A^* and hence relax the short-run implementability constraint). Therefore, the planner has an incentive to let interest rates be higher and to provide more insurance to the domestic economy. Overall, this provides an intuition of why optimal r is decreasing in δ (part (C) of the proposition).

Finally, in a closed economy ($A^* = 0$), the planner does not run costs from issuing debt to foreigners, thus it will always fully insure the domestic economy, setting $\beta(1+r) = 1$ and equalizing consumption between poor and rich agents, regardless of δ.

RoW demand elasticity. In the two-period model of Section 3, a more elastic RoW demand requires higher optimal interest rate (since the planner has less margin to extract monopoly rents). In the infinite horizon model with a general discount factor, the effect of higher elasticity is instead ambiguous and depends on the value of δ. If $1 < \delta(1+r)$, more elastic demand reduces the optimal RSS interest rate (for a given τ). The opposite occurs for $1 > \delta(1+r)$. See Proposition A.3.

Intuitively, when δ is high, an increase in A^* (following higher r) generates a long-run cost for the planner. This cost is increasing in the elasticity of A^*. Hence, for higher elasticity, the planner will optimally set a lower interest rate. On the contrary, for low δ, the increase in A^* may turn out to benefit the planner (since the planner cares more about the short-run benefit of relaxing the implementability constraint). This benefit is increasing in the elasticity ξ^*, in turn pushing the planner to set higher interest rates.

4.4.2 Characterization of the Ramsey Steady State

As described in the previous section, the RSS is characterized by the necessary condition (28) and the implementability constraint (29). These two equations can be rearranged to express the RSS tax rate as a function of the interest rate:

$$\tau = 1 - \frac{(1+\delta)(1+\beta)(1+r)}{\delta[1-\beta(1+r)]} A^*(r)\Big(\delta - [1-\delta(1+r)]\xi^*(r)\Big),$$
$$\tau = \frac{r[\beta + (1+\beta)A^*(r)]}{1+\beta(1+r)}.$$

It is possible to plot these relations in $r - \tau$ space. Figure 1 shows an illustrative example.[20]

[20] I assume that foreign demands follows:

$$A^*(r) = \alpha\left(1 - \frac{1}{(1+r)\beta^*}\right)^{\gamma},$$

meaning that $A^*(r) = 0$ if $1 + r \leq 1/\beta^*$, where β^* captures RoW discount factor, with $\beta^* > \beta$. The elasticity of foreign demand is:

$$\xi^* = \gamma \frac{1}{[(1+r)\beta^* - 1]} \frac{1}{(1+r)}.$$

I calibrate the model parameters to the following values: $\beta = 0.92, \beta^* = 0.98, \alpha = 100, \gamma = 2, \delta = \beta$.

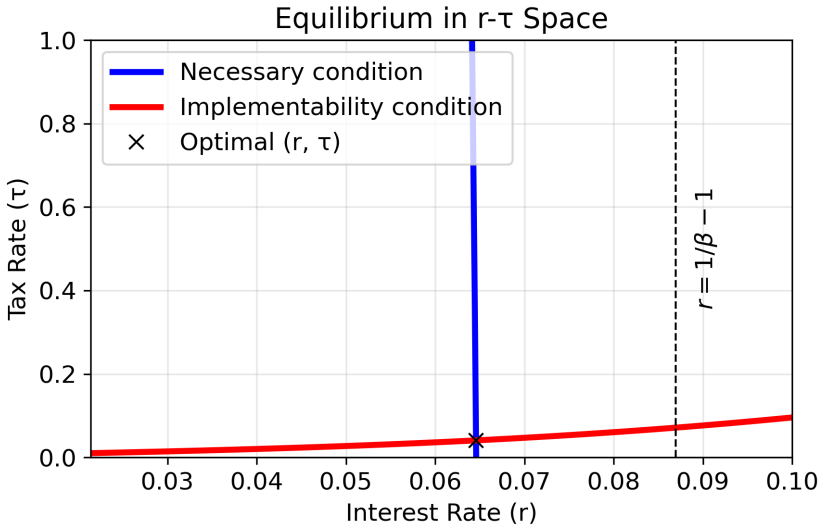

FIGURE 1: CHARACTERIZATION OF THE RSS

The red line is the implementability condition (29). According to it, taxes are increasing in the interest rate since higher taxes are required to finance higher interest payments on the debt. On the contrary, the blue line plots the necessary condition (28), for which higher interest rates are associated with lower τ (see Proposition A.4). The intersection between the two lines determines the RSS (r, τ).

4.4.3 Effects of δ

As discussed in Section 4.4.1, the RSS depends critically on the social discount factor δ. Figure 2 plots the main RSS variables as a function of δ. Red lines represent the optimal solution of equations (28), (29). Green lines represent instead the solution if the planner were able to achieve the first best of full domestic insurance and optimal international rents extraction. This latter case can be achieved by issuing two separate debt instruments: domestic debt that can only be bought by domestic agents and foreign debt that can only be bought by foreign agents. See details in Section 5.

The top-left panel plots the RSS interest rate r as function of δ (red line $r(PI)$). The planner can instead achieve the first best by issuing domestic-only debt and setting the interest rate at the full insurance level $r(FI) = 1/\beta - 1$ (black dash line). In the first best, it will also issue foreign-only debt paying an interest rate of $r^*(FI)$ (green line). It is possible to notice that the interest rate $r(PI)$ (optimal in the partial insurance case) lies in between $r(FI)$ and $r^*(FI)$. This shows how the planner balances the two motives of domestic liquidity provision and optimal international rents extraction. Finally, the partial insurance interest rate is decreasing in δ, as discussed in Section 4.4.1.

The top-right panel plots the RSS tax rate τ. The tax rate in the two-instrument full insurance case (green line) is higher than the partial insurance one (red line). This follows from the fact that in this example the full insurance case interest payments on domestic debt are much higher, and need to be financed with higher tax rates. Taxes are also decreasing in δ since lower interest rates require the government to raise lower tax revenues.

The bottom-left panel plots the levels of debt held by domestic (continuous lines) and foreign

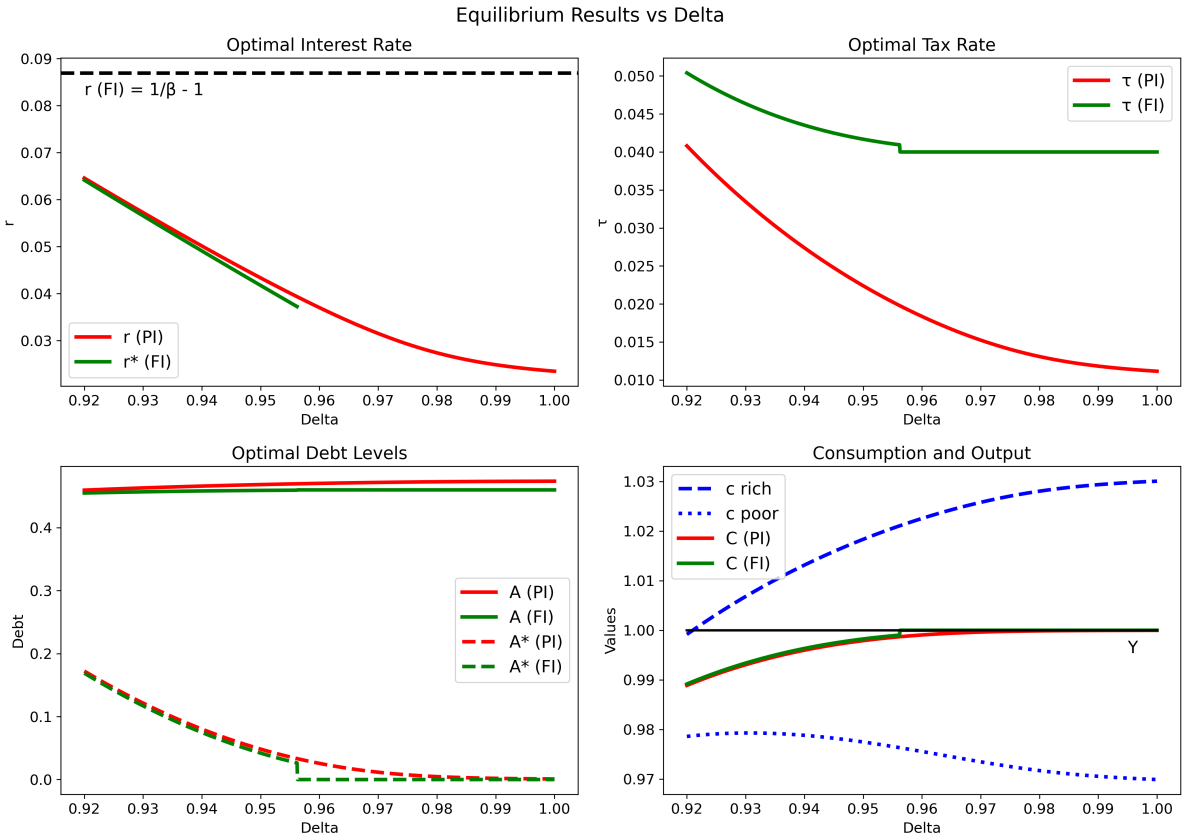

FIGURE 2: COMPARATIVE STATICS FOR δ

(dash lines) agents in the one-instrument partial insurance case (red lines) and the two-instrument first best case (green lines). With higher δ the planner reduces interest rates on foreign debt to cut the amount of borrowing from the RoW. Finally, for sufficiently high δ the planner would like to issue zero debt to foreigners in the first best (since it puts high weight on interest payments to the RoW), as can be seen by the green dash line.

The bottom-right panel plots aggregate consumption (continuous line) in the one-instrument partial insurance case (red lines) and the two-instruments first best case (green lines). With low δ the planner tends to hold negative net foreign assets in the steady state, hence consumption is lower than output (black line). The blue lines plot consumption of poor and rich agents. Inequality is increasing in δ, consistently with the fact that interest rates are lower in δ (and hence the planner provides less insurance to the domestic economy).

Overall, the infinite horizon model confirms and extends the insights from the two-period model. The fundamental trade-off between domestic liquidity provision and interest rate manipulation persists in the long run. The optimal policy generally involves partial insurance rather than full insurance, with the planner strategically restricting domestic liquidity to extract rents from foreign investors. The degree of this trade-off depends on the social discount factor, with more patient planners (higher δ) focusing less on domestic insurance and more on extracting rents from abroad.

5 Two instruments

In this section I show that if the planner is able to issue two debt instruments separately for home and foreign agents, then it can achieve the first best allocation of full insurance and optimal interest rate manipulation.

5.1 Problem setup

I consider the same infinite horizon model of Section 4, with the only difference that the planner issues two debt instruments: home debt B_t and foreign debt B_t^*. Domestic debt pays a risk-free rate r_{t+1} while foreign debt pays a risk-free rate r_{t+1}^*. The government prevents home agents from buying foreign debt and RoW from buying home debt. Under these assumptions, the household problem is unchanged with respect to Section 4. Instead, I can rewrite RoW asset demand as a function of the interest rate r_{t+1}^*: $A_t^*(r_{t+1}^*)$. Finally, the government budget constraint now reads:

$$A_t + A_t^* = (1 + r_t)A_{t-1} + (1 + r_t^*)A_{t-1}^* - \tau_t y_t,$$

since the government issues $A_t(\tau_t) + A_t^*(r_{t+1}^*)$ new debt and repays home debt $(1 + r_t)A_{t-1}$ and foreign debt $(1 + r_t^*)A_{t-1}^*$.

In the partial insurance case, domestic aggregate debt demand is only function of the tax rate, as discussed in Section 4.1.2:

$$A_t(\tau_t) = \frac{\beta}{1+\beta} y_t(1 - \tau_t).$$

Therefore, the planner's implementability constraint is:

$$A_t(\tau_t) + A_t^*(r_{t+1}^*) = (1 + r_t)A_{t-1}(\tau_{t-1}) + (1 + r_t^*)A_{t-1}^*(r_t^*) - \tau_t y_t. \tag{30}$$

Overall, the Ramsey planner solves the following problem:

$$\begin{aligned}
\max_{\{\tau_t, r_{t+1}, r_{t+1}^*\}} \quad & \sum_{t=0}^{\infty} \delta^t [u(c_{r,t}) + u(c_{p,t})] \\
\text{s.t.} \quad & c_{r,t} = \frac{1}{1+\beta} 2y_t(1 - \tau_t), \\
& c_{p,t} = (1 + r_t)2A_{t-1}(\tau_{t-1}), \\
& A_t(\tau_t) = \frac{\beta}{1+\beta} y_t(1 - \tau_t), \\
& A_t(\tau_t) + A_t^*(r_{t+1}^*) = (1 + r_t)A_{t-1}(\tau_{t-1}) + (1 + r_t^*)A_{t-1}^*(r_t^*) - \tau_t y_t.
\end{aligned}$$

5.2 Characterization of the Ramsey problem

Solving the Ramsey problem gives the following FOCs:[21]

$$\begin{aligned}
[\tau_t]: \quad & 2u'(c_{r,t}) - \lambda_t + \beta\delta(1 + r_{t+1})[2u'(c_{p,t+1}) - \lambda_{t+1}] = 0, \\
[r_{t+1}]: \quad & 2u'(c_{p,t}) = \lambda_t, \\
[r_{t+1}^*]: \quad & \lambda_t \xi_t^* - \delta\lambda_{t+1}\left[1 + (1 + r_{t+1}^*)\xi_t^*\right] = 0.
\end{aligned}$$

where $\xi_t^* := \frac{\partial \log A_t^*}{\partial r_{t+1}^*}$ is the usual elasticity of RoW asset demand. I can then state the following proposition.

[21] See Appendix C.1.

Proposition 2. *If the planner can issue two separate debt instruments and discriminate domestic and RoW bond holders, it can achieve the first best of full domestic insurance:*

$$u'(c_{r,t}) = u'(c_{p,t}) = \lambda_t, \tag{31}$$

and optimal foreign interest rate manipulation:

$$C_t^{-1}\xi_t^* = \delta C_{t+1}^{-1}\left[1 + (1 + r_{t+1}^*)\xi_t^*\right] \tag{32}$$

Proof. See Appendix C.2. □

As a consequence of full insurance $c_{r,t} = c_{p,t} = C_t$, and aggregate consumption is determined jointly by equation (32) and the country resource constraint:

$$C_t = y_t + A_t^*(r_{t+1}^*) - (1 + r_t^*)A_{t-1}^*(r_t^*).$$

Notice that it is possible to rewrite (32) as:

$$\begin{aligned} C_t^{-1} &= \delta(1 + r_{t+1}^*)C_{t+1}^{-1}\left[1 + [\xi_t^*(1 + r_{t+1}^*)]^{-1}\right] \\ \implies 1 + r_{t+1}^* &= \frac{1}{\delta}\frac{C_{t+1}}{C_t}\frac{1}{1 + [\xi_t^*(1 + r_{t+1}^*)]^{-1}} \end{aligned}$$

The equation above coincides with the first-best optimal rent extraction Euler equation (3) for a representative agent economy when the planner can discriminate between domestic and foreign agents and internalizes how interest rates respond to debt issuance. In equation (3), Γ_t was defined as the elasticity of the interest rate to debt:

$$\Gamma_t = \frac{\partial r_t}{\partial \log A_t^*}\frac{1}{1 + r_t} = \frac{1}{\xi_t^*}\frac{1}{1 + r_t}.$$

Therefore I can re-express foreign optimal interest rate as:

$$(1 + r_{t+1}^*)(1 + \Gamma_t) = \frac{1}{\delta}\frac{C_{t+1}}{C_t}$$

that coincides with equation (3) if $\delta = \beta$.

Finally, the domestic interest rate is determined by the Euler equation with respect to domestic bonds. Since both agents are on their Euler equation, it is possible to aggregate individual Euler equations and get:

$$1 + r_{t+1} = \frac{1}{\beta}\frac{C_{t+1}}{C_t}.$$

Moreover, domestic interest rate is greater than foreign one:

$$1 + r_{t+1}^* = \frac{1}{\beta}\frac{C_{t+1}}{C_t}\underbrace{\frac{\beta}{\delta}}_{\leq 1}\underbrace{\frac{1}{1 + [\xi_t^*(1 + r_{t+1}^*)]^{-1}}}_{\leq 1} \leq \frac{1}{\beta}\frac{C_{t+1}}{C_t} = 1 + r_{t+1}.$$

Intuitively, in the first best, the planner allows a sufficiently a high return to domestic agents to fully insure the domestic economy, but depresses the interest rate on foreign debt, to reduce interest payments to the RoW.

5.2.1 Characterization of the RSS

Turning to the Ramsey Steady State, conditions (31) and (32) become:

$$c_r = c_p = C \implies r = \beta^{-1} - 1,$$
$$[1 - \delta(1 + r^*)]\xi^*(r^*) - \delta = 0.$$

The second equation determines r^*. Notice, however that for high values of δ there may be no solution and the planner would hence set $A^* = 0$. This is shown in the bottom-left panel of Figure 2 where the green dash line goes to zero for high values of δ. Intuitively, for high δ, the planner cares more about long-run losses from higher interest payments on foreign debt. Hence, it would prefer to set $A^* = 0$. Moreover, r^* is decreasing in δ:

$$\frac{dr}{d\delta} = \frac{(1 + r^*)\xi^* + 1}{-\delta\xi^* + [1 - \delta(1 + r^*)]\frac{\partial \xi^*}{\partial r^*}} = \frac{(1 + r^*)\xi^* + 1}{-\delta\xi^* + \delta\frac{\partial \xi^*/\xi^*}{\partial r^*}} < 0 \quad \text{if } \frac{\partial \xi^*}{\partial r^*} \leq 0$$

Finally, the implementability condition reads:

$$rA + r^* A^*(r^*) = \tau$$

and determines the tax rate and level of debt. However, for the equilibrium to be consistent with full insurance, the level of domestic debt must be set above some minimum level A_{min}:[22]

$$A \geq A_{min} := \frac{\beta}{1 + \beta}(1 - \tau), \quad \tau \leq 1.$$

For instance, if the planner sets domestic debt to its minimum level consistent with full insurance (i.e., A_{min}), the tax rate equals:

$$\tau = \frac{1 - \beta + (1 + \beta)r^* A^*(r^*)}{2}.$$

Figure 2 plots with green lines the equilibrium values for the case in which the planner issues the two separate debt instruments, assuming that domestic debt is set to this minimum level A_{min}.

6 Conclusion

This paper has studied which is the optimal level of government debt and what are its determinants. In particular, debt benefits the domestic economy since it provides liquidity and it partially insures domestic agents from idiosyncratic risk. Moreover, in an open economy, the government has an incentive to under-issue government debt in such a way to reduce interest payments to foreign residents.

I have shown that a Ramsey planner that maximizes utilitarian welfare tradeoffs liquidity provision and interest rate manipulation to extract rents from the rest of the world. In fact, a planner that fully insures the domestic economy but cannot prevent foreign agents from buying the domestic debt, loses the ability to control interest rates, therefore foregoing rent extraction. Overall, the planner can improve welfare if it only partially insures the domestic economy, since under-provision of debt allows to cut interest rates and payments to the rest of the world.

[22] See Appendix C.2.

Then, I have characterized the trade-off between liquidity provision and interest rate manipulation in a stylized two-agents model with alternating endowments over time (both in a two-period and in an infinite horizon model). In the infinite horizon version, I characterize the Ramsey steady state for different planner's discount factors. Importantly, the openness of the economy makes the planner deviate from full insurance. The under-provision of liquidity is stronger the more inelastic is foreign demand for domestic debt (if the planner discount factor is relatively low and close to the households' one). If the planner gives more weight to long-run outcomes, it will set lower interest rates since it takes into account the losses from steady-state interest payments to the rest of the world.

Finally, I have shown that it is possible to decentralize the first best of full domestic insurance and optimal interest rate manipulation by issuing two separate debt instruments: domestic debt that can only be bought by domestic agents and foreign debt that can only be bought by foreign agents.

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APPENDIX

A Proofs for the two-period model

Proposition A.2. *The aggregate asset demand function* (16)

$$A_t(r_{t+1}, B_t) = \begin{cases} \frac{\beta}{1+\beta}[y_t - (1+r_t)B_{t-1}] + \frac{\beta}{1+\beta}B_t & \text{if } B_t \le \hat{B}_t, \\ \frac{1}{1+\beta}\left(\beta[y_t - (1+r_t)A^*_{t-1}] - \frac{y_{t+1}}{1+r_{t+1}}\right) + B_t & \text{if } B_t > \hat{B}_t. \end{cases}$$

is a continuous function of B_t.

Proof. Continuity for $B_t < \hat{B}_t$ or $B_t > \hat{B}_t$ is obvious. I show continuity at $\hat{B}_t$ by rewriting the asset demand in case of full insurance ($B_t > \hat{B}_t$) in the following way:

$$\frac{1}{1+\beta}\left(\beta[y_t - (1+r_t)A^*_{t-1}] - \frac{y_{t+1}}{1+r_{t+1}}\right) + B_t =$$
$$= \frac{\beta}{1+\beta}[y_t - (1+r_t)B_{t-1}] + \frac{\beta}{1+\beta}B_t + \frac{1}{1+\beta}\Bigg[B_t - \underbrace{\left(\frac{y_{t+1}}{1+r_{t+1}} - \beta(1+r_t)A_{t-1}\right)}_{=\hat{B}_t}\Bigg].$$

When taking the limit of $B_t \to \hat{B}_t$ I get:

$$\lim_{B_t \uparrow \hat{B}_t} A_t(r_{t+1}, B_t) = \lim_{B_t \downarrow \hat{B}_t} A_t(r_{t+1}, B_t) = A_t(r_{t+1}, \hat{B}_t).$$

that shows continuity of the function at $\hat{B}_t$. □

A.1 Planner problem

A more complete formulation of the planner problem is the following:

$$\begin{aligned} \max_{B_t} \quad & \sum_{s=0}^{1} \beta^s [u(c_{r,t+s}) + u(c_{p,t+s})] \\ \text{s.t.} \quad & \frac{1}{2}c_{r,t} + \frac{1}{2}c_{p,t} = y_t + A^*_t(r_{t+1}) - (1+r_t)A^*_{t-1}, \\ & \frac{1}{2}c_{r,t+1} + \frac{1}{2}c_{p,t+1} = y_{t+1} - (1+r_{t+1})A^*_t(r_{t+1}), \\ & c_{p,t} = \begin{cases} (1+r_t)2A_{t-1} & \text{if } B_t \le \hat{B}_t \\ \frac{1}{1+\beta}\left[(1+r_t)2A_{t-1} + \frac{2y_{t+1}}{1+r_{t+1}} - 2B_t\right] & \text{if } B_t > \hat{B}_t, \end{cases} \\ & c_{p,t+1} = \begin{cases} (1+r_{t+1})2[B_t - A^*_t(r_{t+1})] & \text{if } B_t \le \hat{B}_t \\ \frac{\beta(1+r_{t+1})}{1+\beta}2\left[y_t + B_t - (1+r_t)B_{t-1}\right] & \text{if } B_t > \hat{B}_t, \end{cases} \\ & r_{t+1} = \mathcal{R}_t(B_t). \end{aligned}$$

The time t poor agent consumption does not depend on B_t in the partial insurance case. However, if the planner sets $B_t > \hat{B}_t$ then $c_{p,t}$ is decreasing in B_t. When the poor agent is on its Euler

equation, its consumption is given by the following:

$$\begin{aligned} c_{p,t} &= \frac{1}{1+\beta}\left[(1+r_t)2A_{t-1} + \frac{2y_{t+1}(1-\tau_{t+1})}{1+r_{t+1}}\right] \\ &= \frac{1}{1+\beta}\left[(1+r_t)2A_{t-1} + \frac{2y_{t+1}}{1+r_{t+1}} - 2B_t\right] \end{aligned}$$

where we have used the second period government budget constraint (5) to establish the second equality. The poor agent cuts here current consumption in anticipation of future taxes (increasing in the level of debt). Similarly, in the full insurance case, the rich agent future consumption is given by:

$$\begin{aligned} c_{p,t+1} &= \frac{\beta(1+r_{t+1})}{1+\beta}2y_t(1-\tau_t) \\ &= \frac{\beta(1+r_{t+1})}{1+\beta}2\left[y_t + B_t - (1+r_t)B_{t-1}\right] \text{ if } B_t > \hat{B}_t. \end{aligned}$$

Since $\mathcal{R}(B_t)$, $c_{p,t}$, and $c_{p,t+1}$ are continuous functions of B_t, we can argue that welfare $W(B_t)$ is also continuous in B_t. However, its derivative is generally not continuous at $\hat{B}_t$. Our proof of optimality of partial insurance relies on showing that $\lim_{B_t \uparrow \hat{B}_t} W'(B_t) < 0$ while $W'(B_t) < 0$ for $B_t > \hat{B}_t$.

Overall, in case the planner sets $B_t \geq \hat{B}_t$, the relevant problem can be rewritten as:

$$\begin{aligned} \max_{B_t} \quad & \sum_{s=0}^{1} \beta^s [u(c_{r,t+s}) + u(c_{p,t+s})] \\ \text{s.t.} \quad & \frac{1}{2}c_{r,t} + \frac{1}{2}c_{p,t} = y_t + A_t^*(r_{t+1}^{FI}) - (1+r_t)A_{t-1}^*, \\ & \frac{1}{2}c_{r,t+1} + \frac{1}{2}c_{p,t+1} = y_{t+1} - (1+r_{t+1}^{FI})A_t^*(r_{t+1}^{FI}), \\ & c_{p,t} = \frac{1}{1+\beta}\left[(1+r_t)2A_{t-1} + \frac{2y_{t+1}}{1+r_{t+1}^{FI}} - 2B_t\right], \\ & c_{p,t+1} = \frac{\beta(1+r_{t+1}^{FI})}{1+\beta}2\left[y_t + B_t - (1+r_t)B_{t-1}\right]. \end{aligned}$$

A.2 Solution of the planner's problem

Here I derive $W'(B)$. I define W_p, W_r as the value function of the time t poor and rich agent respectively.

$$\begin{aligned} W_p(\tau_{t+1}, r_{t+1}) := \max_{c_{p,t}, c_{r,t+1}, a_{p,t}} \quad & \log(c_{p,t}) + \beta\log(c_{r,t+1}) \\ \text{s.t.} \quad & c_{p,t} + a_{p,t} = (1+r_t)A_{t-1},\ a_{p,t} \geq 0, \\ & c_{r,t+1} = 2y_{t+1}(1-\tau_{t+1}) + (1+r_{t+1})a_{p,t}, \\ W_r(\tau_t, r_{t+1}) := \max_{c_{r,t}, c_{p,t+1}, a_{r,t}} \quad & \log(c_{r,t}) + \beta\log(c_{p,t+1}) \\ \text{s.t.} \quad & c_{r,t} + a_{r,t} = 2y_t(1-\tau_t),\ a_{r,t} \geq 0, \\ & c_{p,t+1} = (1+r_{t+1})a_{r,t}. \end{aligned}$$

Partial insurance. Consider first the partial insurance case $B_t \leq \hat{B}_t$. Changes in debt do not affect $c_{p,t}$ but instead affect $c_{r,t+1}$ since the $t+1$ taxes will increase to repay the debt. Hence the welfare of the poor agent changes:

$$\begin{aligned}\frac{dW_p}{dB_t} &= -\beta u'(c_{r,t+1}) \cdot 2y_{t+1}\frac{d\tau_{t+1}}{dB_t} \\ &= -2\beta u'(c_{r,t+1}) \cdot \left[(1+r_{t+1}) + B_t\frac{dr_{t+1}}{dB_t}\right].\end{aligned}$$

where we have used the government budget constraint for:

$$y_{t+1}\frac{d\tau_{t+1}}{dB_t} = (1+r_{t+1}) + B_t\frac{dr_{t+1}}{dB_t}.$$

For the time t rich agent, its value function can be re-expressed as:

$$\begin{aligned}W_r(\tau_t, r_{t+1}) := \max_{c_{r,t}, c_{p,t+1}} \quad & u(c_{r,t}) + \beta u(c_{p,t+1}) \\ \text{s.t.} \quad & c_{r,t} + \frac{c_{p,t+1}}{1+r_{t+1}} = 2y_t(1-\tau_t).\end{aligned}$$

Let λ_r be the Lagrange multiplier and notice that it must satisfy $\lambda_r = u'(c_{r,t}) = \beta(1+r_{t+1})u'(c_{p,t+1})$. Using the Envelope theorem, the change in rich agent's welfare is:

$$\begin{aligned}\frac{dW_r}{dB_t} &= \lambda_r\left[-2y_t\frac{d\tau_t}{dB_t} + \frac{c_{p,t+1}}{(1+r_{t+1})^2}\frac{dr_{t+1}}{dB_t}\right] \\ &= \lambda_r\left[2 + \frac{c_{p,t+1}}{(1+r_{t+1})^2}\frac{dr_{t+1}}{dB_t}\right]\end{aligned}$$

since

$$y_t\frac{d\tau_t}{dB_t} = -1.$$

Overall, total change in welfare is:

$$\begin{aligned}\frac{1}{2}\frac{dW}{dB_t} &= \frac{1}{2}\left[\frac{dW_p}{dB_t} + \frac{dW_p}{dB_t}\right] \\ &= -\beta(1+r_{t+1})u'(c_{r,t+1}) \cdot [1+\Gamma_t] + \beta(1+r_{t+1})u'(c_{p,t+1})\left[1 + \frac{1}{2}\frac{c_{p,t+1}}{1+r_{t+1}}\frac{1}{B_t}\Gamma_t\right] \\ &= -\beta(1+r_{t+1})u'(c_{r,t+1}) \cdot [1+\Gamma_t] + \beta(1+r_{t+1})u'(c_{p,t+1})\left[1+\Gamma_t - s_t^*\Gamma_t\right] \\ &= \beta[u'(c_{p,t+1}) - u'(c_{r,t+1})] \cdot (1+r_{t+1})(1+\Gamma_t) - \beta u'(c_{p,t+1}) \cdot (1+r_{t+1})s_t^*\Gamma_t\end{aligned}$$

where the second equality uses also the definition of $\Gamma_t = (dr_{t+1}/(1+r_{t+1}))/(d\log B_t)$ while the third equality uses:

$$\frac{c_{p,t+1}}{2} = (1+r_{t+1})B_t(1-s_t^*), \quad s_t^* = \frac{A_t^*}{B_t}.$$

Full insurance. For the full insurance case $r_{t+1} = r_{t+1}^{FI}$ is fixed, but $c_{p,t}$ is decreasing in B_t. Therefore we have:

$$\begin{aligned}\frac{dW_p}{dB_t} &= -\lambda_p \cdot \frac{2y_{t+1}}{1+r_{t+1}} \frac{d\tau_{t+1}}{dB_t} \\ &= -\beta(1+r_{t+1})u'(c_{r,t+1}), \\ \frac{dW_r}{dB_t} &= -\lambda_r 2y_t \frac{d\tau_t}{dB_t} \\ &= 2\beta(1+r_{t+1})u'(c_{p,t+1}),\end{aligned}$$

where $\lambda_p = u'(c_{p,t}) = \beta(1+r_{t+1})u'(c_{r,t+1})$ is the Lagrange multiplier of the time t poor agent (that is on her Euler equation in the full insurance case). Overall, I get:

$$\frac{1}{2}\frac{dW}{dB_t} = \beta[u'(c_{p,t+1}) - u'(c_{r,t+1})] \cdot (1+r_{t+1}).$$

A.3 Intuitive decomposition

To obtain the intuitive decomposition (21) notice that:

$$\begin{aligned}\beta[u'(c_{p,t+1}) - u'(c_{r,t+1})] \cdot (1+r_{t+1}) &= [\beta(1+r_{t+1})u'(c_{p,t+1}) - u'(c_{r,t})] + \\ &+ [u'(c_{p,t}) - \beta(1+r_{t+1})u'(c_{r,t+1})] + [u'(c_{r,t}) - u'(c_{p,t})] \\ &= [u'(c_{p,t}) - \beta(1+r_{t+1})u'(c_{r,t+1})] + [u'(c_{r,t}) - u'(c_{p,t})]\end{aligned}$$

since $u'(c_{r,t}) = \beta(1+r_{t+1})u'(c_{p,t+1})$. Therefore I get:

$$\begin{aligned}\frac{dW_t}{dB_t} &= [u'(c_{p,t}) - \beta(1+r_{t+1})u'(c_{r,t+1})](1+\Gamma_t) + [u'(c_{r,t}) - u'(c_{p,t})](1+\Gamma_t) \\ &+ u'(c_{r,t}) - \beta(1+r_{t+1})u'(c_{p,t+1})(1+s_t^*\Gamma_t).\end{aligned}$$

Studying the signs. For $B_t < \hat{B}_t$ I have that poor agents are partially insured hence $u'(c_{p,t}) > \beta(1+r_{t+1})u'(c_{r,t+1})$. Since rich agents are on their Euler equation I also have that: $u'(c_{r,t}) - \beta(1+r_{t+1})u'(c_{p,t+1})(1+s_t^*\Gamma_t) = -u'(c_{p,t+1})s_t^*\Gamma_t < 0$. Finally, turning to the residual inequality term $[u'(c_{r,t}) - u'(c_{p,t})](1+\Gamma_t)$, it is negative iff $c_{r,t} > c_{p,t}$. Recall that:

$$\begin{aligned}c_{r,t} &= \frac{2}{1+\beta}y_t(1-\tau_t) = \frac{2}{1+\beta}[y_t + B_t - (1+r_t)B_{t-1}], \\ c_{p,t} &= \begin{cases}(1+r_t)2A_{t-1} & \text{if } B_t \le \hat{B}_t \\ \frac{1}{1+\beta}\left[(1+r_t)2A_{t-1} + \frac{2y_{t+1}}{1+r_{t+1}} - 2B_t\right] & \text{if } B_t > \hat{B}_t.\end{cases}\end{aligned}$$

Hence $c_{r,t}$ is increasing in B_t while $c_{p,t}$ is constant and decreasing in B_t for $B_t > \hat{B}_t$. This implies that the term $[u'(c_{r,t}) - u'(c_{p,t})]$ is decreasing in B_t. Under the following assumption $c_{r,t} > c_{p,t}$ in a neighborhood of the full insurance level of debt ($\hat{B}_t$), meaning that pure redistribution term is less or equal than zero at $\hat{B}_t$.

Assumption A.2. *Initial assets of the time t poor agent are sufficiently small:*

$$(1+r_t)A_{t-1} \le \frac{1}{2}\left[y_t - (1+r_t)A_{t-1}^* + A_t^*(r_{t+1}^{FI})\right]$$

In fact, for $B_t = \hat{B}_t$:

$$
\begin{aligned}
c_{r,t} \geq c_{p,t} &\iff \frac{1}{1+\beta}[y_t + B_t - (1+r_t)B_{t-1}] \geq (1+r_t)A_{t-1} \\
&\iff \frac{1}{1+\beta}\Big[y_t + \underbrace{\frac{y_{t+1}}{1+r^{FI}_{t+1}} - \beta(1+r_t)A_{t-1}}_{=\hat{B}_t} - \underbrace{(1+r_t)B_{t-1}}_{=(1+r_t)(A_{t-1}+A^*_{t-1})}\Big] \geq (1+r_t)A_{t-1} \\
&\iff \frac{1}{1+\beta}\Big[y_t + \frac{y_{t+1}}{1+r^{FI}_{t+1}} - (1+r_t)A^*_{t-1}\Big] \geq 2(1+r_t)A_{t-1} \\
&\iff y_t - (1+r_t)A^*_{t-1} + A^*_t(r^{FI}_{t+1}) \geq 2(1+r_t)A_{t-1}
\end{aligned}
$$

where the last equality can be obtained substituting:

$$
1 + r^{FI}_{t+1} = \frac{y_{t+1}}{\beta[y_t - (1+r_t)A^*_{t-1}] + (1+\beta)A^*_t(r^{FI}_{t+1})}.
$$

Therefore, as $B_t \uparrow \hat{B}_t$, $c_{r,t} \geq c_{p,t}$ and the pure redistribution terms operates in a direction opposite to the pure insurance term. More debt further redistributes income to the time t rich agents, hence the planner is willing not to increase debt too much. Notice also that assumption 1 can be reformulated as:

$$
(1+r_t)A_{t-1} \leq y_t - (1+r_t)A^*_{t-1} + A^*_t(r^{FI}_{t+1})(1+\beta^{-1}),
$$

meaning that assumption A.2 imposes a stronger restriction than assumption 1 (equivalently assumption A.2 is sufficient for 1).

Optimality of partial insurance. Under Assumption A.2, partial insurance is optimal, since $W'(\hat{B}_t) < 0$ and $W'(B_t) < 0$ for $B_t > \hat{B}_t$ since the redistribution term is decreasing in B_t for $B_t > \hat{B}_t$.

A.4 Closed economy and inelastic asset demand

Here I show that in the case of a perfectly inelastic foreign asset demand only full insurance levels of debt can be supported in equilibrium. Let $A^*_t \geq 0$ be constant. The closed economy is a special case of $A^*_t = 0$.

First I rewrite the minimum full insurance level of debt as:

$$
\begin{aligned}
\hat{B}_t &= \frac{y_{t+1}}{1+r^{FI}_{t+1}} - \beta(1+r_t)A_{t-1} \\
&= \beta[y_t - (1+r_t)A^*_{t-1}] + (1+\beta)A^*_t - \beta(1+r_t)A_{t-1} \\
&= \beta[y_t - (1+r_t)B_{t-1}] + (1+\beta)A^*_t
\end{aligned}
$$

Consider aggregate domestic asset demand (16).

$$
A_t(r_{t+1}, B_t) = \begin{cases} \frac{\beta}{1+\beta}[y_t - (1+r_t)B_{t-1}] + \frac{\beta}{1+\beta}B_t & \text{if } \; B_t \leq \hat{B}_t, \\ \\ \frac{1}{1+\beta}\left(\beta[y_t - (1+r_t)A^*_{t-1}] - \frac{y_{t+1}}{1+r_{t+1}}\right) + B_t & \text{if } \; B_t > \hat{B}_t. \end{cases}
$$

Suppose the planner choses a partial insurance level of debt $B_t < \hat{B}_t$. Then the asset market clearing reads:

$$
\begin{aligned}
B_t &= \frac{\beta}{1+\beta}[y_t - (1+r_t)B_{t-1}] + \frac{\beta}{1+\beta}B_t + A_t^* \\
\implies B_t &= \beta[y_t - (1+r_t)B_{t-1}] + (1+\beta)A_t^* = \hat{B}_t
\end{aligned}
$$

that is a contradiction.

B Proofs for the infinite horizon model

B.1 FOCs of the Ramsey problem

I derive the FOCs of the planner's problem:

$$
\begin{aligned}
\max_{\{\tau_t, r_{t+1}\}_{t=0}^{\infty}} \quad & \sum_{t=0}^{\infty} \delta^t [u(c_{r,t}) + u(c_{p,t})] \\
\text{s.t.} \quad & c_{p,t} = (1+r_t)2A_{t-1}(\tau_{t-1}), \\
& c_{r,t} = \frac{1}{1+\beta}2(1-\tau_t)y_t, \\
& A_t(\tau_t) = \frac{\beta}{1+\beta}y_t(1-\tau_t), \\
& A_t(\tau_t) + A_t^*(r_{t+1}) = (1+r_t)[A_{t-1}(\tau_{t-1}) + A_{t-1}^*(r_t)] - \tau_t y_t.
\end{aligned}
$$

The FOCs are:

$$
\begin{aligned}
\tau_t: \quad & 0 = -u'(c_{r,t})\frac{1}{1+\beta}2y_t + 2\delta u'(c_{p,t+1})(1+r_{t+1})\frac{\partial A_t}{\partial \tau_t} + \lambda_t\left[\frac{\partial A_t}{\partial \tau_t} + y_t\right] - \delta\lambda_{t+1}(1+r_{t+1})\frac{\partial A_t}{\partial \tau_t} \\
\implies & 0 = -u'(c_{r,t})\frac{1}{1+\beta}2y_t - 2\delta u'(c_{p,t+1})(1+r_{t+1})\frac{\beta}{1+\beta}y_t + \lambda_t\left[-\frac{\beta}{1+\beta}y_t + y_t\right] + \delta\lambda_{t+1}(1+r_{t+1})\frac{\beta}{1+\beta}y_t \\
\implies & 0 = -u'(c_{r,t})2 - 2\beta\delta u'(c_{p,t+1})(1+r_{t+1}) + \lambda_t + \beta\delta\lambda_{t+1}(1+r_{t+1}) \\
\implies & 0 = (1+\delta)2u'(c_{r,t}) - [\lambda_t + \beta\delta\lambda_{t+1}(1+r_{t+1})] \\
r_{t+1}: \quad & 0 = \delta u'(c_{p,t+1})2A_t - \delta\lambda_{t+1}A_t + \lambda_t\frac{\partial A_t^*}{\partial r_{t+1}} - \delta\lambda_{t+1}\left[A_t^* + (1+r_{t+1})\frac{\partial A_t^*}{\partial r_{t+1}}\right] \\
\implies & 0 = [2u'(c_{p,t+1}) - \lambda_{t+1}]\delta A_t + \lambda_t A_t^*\xi_t^* - A_t^*\delta\lambda_{t+1}[1 + (1+r_{t+1})\xi_t^*]
\end{aligned}
$$

where I have used the Euler equation of rich agents: $\beta(1+r_{t+1})u'(c_{p,t+1}) = u'(c_{r,t})$.

B.2 Characterization of the RSS

B.2.1 Derivation of equation (28)

Consider a converging sequence of output: $\{y_t\}_{t=0}^{\infty} \longrightarrow y = 1$. First, I assume that a Ramsey plan exists, hence it satisfies the first-order conditions (26, 27). Second, I assume that the Ramsey plan converges to a steady state: $\{\tau_t, r_{t+1}\}_{t=0}^{\infty} \longrightarrow (\tau, r)$, that is indeed the RSS. Third, I assume that Lagrange multipliers converge $\{\lambda_t\}_{t=0}^{\infty} \longrightarrow \lambda$.

First notice that assets and consumption policy functions converge:

$$A_t(\tau_t) \longrightarrow A(\tau) = \frac{\beta}{1+\beta}(1-\tau),$$
$$c_{p,t} \longrightarrow c_p = (1+r)2A(\tau),$$
$$c_{r,t} \longrightarrow c_r = \frac{1}{1+\beta}2(1-\tau).$$

Assuming that the sequence of Lagrange multiplier converges, from equation (26) I get:

$$\lambda_t \longrightarrow \lambda = \frac{2(1+\delta)u'(c_r)}{1+\beta\delta(1+r)}.$$

Finally, equation (27) holds in the RSS:

$$\left[u'(c_p) - \frac{\lambda}{2}\right]2\delta A + A^*\Big[1-\delta(1+r)\Big]\lambda\xi^* - \delta A^*\lambda = 0.$$

Rearranging the equation above and substituting λ we get:

$$\left[2\frac{u'(c_p)}{\lambda} - 1\right]\delta A + A^*\Big[1-\delta(1+r)\Big]\xi^* - \delta A^* = 0$$
$$\left[\frac{1+\beta\delta(1+r)}{(1+\delta)}\frac{u'(c_p)}{u'(c_r)} - 1\right]\delta A - A^*\Big(\delta - [1-\delta(1+r)]\xi^*\Big) = 0$$
$$\left[\frac{1+\beta\delta(1+r)}{(1+\delta)}\frac{c_r}{c_p} - 1\right]\delta A - A^*\Big(\delta - [1-\delta(1+r)]\xi^*\Big) = 0$$
$$\left[\frac{1+\beta\delta(1+r)}{(1+\delta)}\frac{\frac{1}{1+\beta}2(1-\tau)}{(1+r)\frac{\beta}{1+\beta}2(1-\tau)} - 1\right]\delta A - A^*\Big(\delta - [1-\delta(1+r)]\xi^*\Big) = 0$$
$$\frac{1-\beta(1+r)}{(1+\delta)\beta(1+r)}\delta A - A^*\Big(\delta - [1-\delta(1+r)]\xi^*\Big) = 0.$$

After substituting A I get equation (28):

$$[1-\beta(1+r)]\frac{\delta}{(1+\beta)(1+r)}(1-\tau) - (1+\delta)A^*\Big(\delta - [1-\delta(1+r)]\xi^*\Big) = 0.$$

B.2.2 Proof of Proposition 1

Assumption A.3. *Assume the following:*

(i) $\beta \geq \sqrt{5}/2 - 1/2$;

(ii) A^* *is twice-differentiable;*

(iii) $\partial^2 A^*/\partial r^2$ *is bounded locally (around* r *solving equation* (28)*):*

$$\frac{\partial^2 A^*}{\partial r^2} \in [-M, M],\ M := \left|\frac{1}{\delta(1+r)-1}\right| \cdot \frac{1}{(1+\delta)(1+\beta)(1+r)}\left[\frac{\delta(1-\tau)}{1+r} + (1+\delta)(1+\beta)(1+r)A^*2\xi^*\delta\right].$$

Proof. I prove the three parts of the proposition.

(A) I can rearrange equation (28) in the following way:

$$[1-\beta(1+r)]\frac{\delta}{(1+\beta)(1+r)}(1-\tau)-(1+\delta)A^*\Big(\delta-[1-\delta(1+r)]\xi^*\Big)=0$$
$$[1-\beta(1+r)]\frac{\delta}{(1+\beta)(1+r)}(1-\tau)-(1+\delta)A^*\Big(\delta-[1-(\beta+\delta-\beta)(1+r)]\xi^*\Big)=0$$
$$[1-\beta(1+r)]\left[\frac{\delta}{(1+\beta)(1+r)}(1-\tau)+(1+\delta)A^*\xi^*\right]=\underbrace{(1+\delta)A^*[\delta+(\delta-\beta)(1+r)\xi^*]}_{>0}$$

As noticed above, the term on the right-hand-side is always positive (if $A^*>0$) since $\delta\geq\beta$. This implies that:

$$\beta(1+r)<1$$

(B) For the closed economy, set $A^*=0$. Equation (28) becomes:

$$[1-\beta(1+r)]\frac{\delta}{(1+\beta)(1+r)}(1-\tau)=0$$

that is solved by $\beta(1+r)=1$.

(C) Consider equation (28), that can be rewritten as:

$$[1-\beta(1+r)]\delta(1-\tau)-(1+\delta)(1+\beta)(1+r)A^*\Big(\delta-[1-\delta(1+r)]\xi^*\Big)=0$$

Take the total differential with respect to $dr, d\delta$:

$$\begin{aligned}0=&dr\Bigg[-\beta\delta(1-\tau)-(1+\delta)(1+\beta)A^*\Big(\delta-[1-\delta(1+r)]\xi^*\Big)\\&-(1+\delta)(1+\beta)(1+r)\frac{\partial A^*}{\partial r}\Big(\delta-[1-\delta(1+r)]\xi^*\Big)\\&-(1+\delta)(1+\beta)(1+r)A^*\Big(\delta\xi^*-[1-\delta(1+r)]\frac{\partial\xi^*}{\partial r}\Big)\Bigg]+\\&+d\delta\Bigg[[1-\beta(1+r)](1-\tau)-(1+\beta)(1+r)A^*\Big(\delta-[1-\delta(1+r)]\xi^*\Big)\\&-(1+\delta)(1+\beta)(1+r)A^*\Big(1+(1+r)\xi^*\Big)\Bigg]\end{aligned}$$

that can be rearranged in the following way. Notice that the first line becomes:

$$\begin{aligned}&-\beta\delta(1-\tau)-(1+\delta)(1+\beta)A^*\Big(\delta-[1-\delta(1+r)]\xi^*\Big)=\\&=\frac{1}{1+r}\Bigg[\underbrace{[1-\beta(1+r)]\delta(1-\tau)-(1+\delta)(1+\beta)(1+r)A^*\Big(\delta-[1-\delta(1+r)]\xi^*\Big)}_{=0}\Bigg]-\frac{\delta(1-\tau)}{1+r}=\\&=-\frac{\delta(1-\tau)}{1+r}\end{aligned}$$

The second and third rows can be rewritten as:

$$
\begin{aligned}
&-(1+\delta)(1+\beta)(1+r)\frac{\partial A^*}{\partial r}\big(\delta-[1-\delta(1+r)]\xi^*\big)-(1+\delta)(1+\beta)(1+r)A^*\Big(\delta\xi^*-[1-\delta(1+r)]\frac{\partial \xi^*}{\partial r}\Big)= \\
&=-(1+\delta)(1+\beta)(1+r)A^*\xi^*\big(\delta-[1-\delta(1+r)]\xi^*\big)-(1+\delta)(1+\beta)(1+r)A^*\Big(\delta\xi^*-[1-\delta(1+r)]\frac{\partial \xi^*}{\partial r}\Big)= \\
&=-(1+\delta)(1+\beta)(1+r)A^*\left[\xi^*\big(\delta-[1-\delta(1+r)]\xi^*\big)+\Big(\delta\xi^*-[1-\delta(1+r)]\frac{\partial \xi^*}{\partial r}\Big)\right]
\end{aligned}
$$

Moreover, notice that:

$$
\frac{\partial \xi^*}{\partial r}=\frac{\partial}{\partial r}\left[\frac{\partial A^*}{\partial r}\frac{1}{A^*}\right]=\frac{\partial^2 A^*}{\partial r^2}\frac{1}{A^*}-\left(\frac{1}{A^*}\frac{\partial A^*}{\partial r}\right)^2=\frac{\partial^2 A^*}{\partial r^2}\frac{1}{A^*}-(\xi^*)^2
$$

Substituting this in the expression above I get:

$$
\begin{aligned}
&=-(1+\delta)(1+\beta)(1+r)A^*\left[\xi^*\big(\delta-[1-\delta(1+r)]\xi^*\big)+\Big(\delta\xi^*-[1-\delta(1+r)]\frac{\partial \xi^*}{\partial r}\Big)\right] \\
&=-(1+\delta)(1+\beta)(1+r)A^*\left[2\xi^*\delta-[1-\delta(1+r)](\xi^*)^2+[1-\delta(1+r)](\xi^*)^2-[1-\delta(1+r)]\frac{\partial^2 A^*}{\partial r^2}\frac{1}{A^*}\right] \\
&=-(1+\delta)(1+\beta)(1+r)A^*\left[2\xi^*\delta-[1-\delta(1+r)]\frac{\partial^2 A^*}{\partial r^2}\frac{1}{A^*}\right]
\end{aligned}
$$

The fourth row becomes:

$$
\begin{aligned}
&[1-\beta(1+r)](1-\tau)-(1+\beta)(1+r)A^*\big(\delta-[1-\delta(1+r)]\xi^*\big)= \\
&\frac{1}{1+\delta}\left[[1-\beta(1+r)](1+\delta)(1-\tau)-(1+\delta)(1+\beta)(1+r)A^*\big(\delta-[1-\delta(1+r)]\xi^*\big)\right]= \\
&\frac{1}{1+\delta}[1-\beta(1+r)](1-\tau)
\end{aligned}
$$

Combining these observations, the total differential can be written as:

$$
\begin{aligned}
0=&dr\underbrace{\left[-\frac{\delta(1-\tau)}{1+r}-(1+\delta)(1+\beta)(1+r)A^*\left[\xi^*\delta-[1-\delta(1+r)]\frac{\partial^2 A^*}{\partial r^2}\frac{1}{A^*}\right]\right.}_{=:\Lambda}+ \\
&+d\delta\underbrace{\left[\frac{1}{1+\delta}[1-\beta(1+r)](1-\tau)-(1+\delta)(1+\beta)(1+r)A^*\big(1+(1+r)\xi^*\big)\right]}_{=:\Xi} \\
\implies \frac{dr}{d\delta}=&\frac{-\Xi}{\Lambda}.
\end{aligned}
$$

The coefficient Ξ is negative for sufficiently high β (it's enough to assume that $\delta \geq \beta \geq \sqrt{2}/2-1/2$) as it can be verified comparing it to the (28). In fact:

$$
\begin{aligned}
&(1+\delta)(1+\beta)(1+r)A^*\big(1+(1+r)\xi^*\big)\geq(1+\delta)(1+\beta)(1+r)A^*\big(\delta-[1-\delta(1+r)]\xi^*\big)= \\
&=[1-\beta(1+r)]\delta(1-\tau)\geq\frac{1}{1+\delta}[1-\beta(1+r)](1-\tau)\implies\Xi<0
\end{aligned}
$$

where the last inequality holds for $\delta \geq \sqrt{2}/2 - 1/2$.

Concerning the coefficient for dr, it is negative as long as the second derivative of A^* is bounded. In particular, the following local (i.e., around the r solving equation (28)) bound for $\partial^2 A^*/\partial r^2$ is needed:

$$[\delta(1+r)-1]\frac{\partial^2 A^*}{\partial r^2} \geq -\frac{1}{(1+\delta)(1+\beta)(1+r)}\left[\frac{\delta(1-\tau)}{1+r} + (1+\delta)(1+\beta)(1+r)A^* 2\xi^*\delta\right]$$

That is implied by assumption A.3, i.e., defining M as:

$$M := \left|\frac{1}{\delta(1+r)-1}\right| \cdot \frac{1}{(1+\delta)(1+\beta)(1+r)}\left[\frac{\delta(1-\tau)}{1+r} + (1+\delta)(1+\beta)(1+r)A^* 2\xi^*\delta\right].$$

the previous inequality holds under:

$$\frac{\partial^2 A^*}{\partial r^2} \in [-M, M].$$

I conclude that $\Lambda < 0$ hence: $\frac{dr}{d\delta} < 0$.

□

B.2.3 Elasticity of foreign demand 1

Proposition A.3. *Under Assumption A.3 (ii) and (iii), an increase in the elasticity of RoW demand ξ^* increases the interest rate according to equation (28) $\iff$ $\delta(1+r) < 1$.*

Proof. Consider equation (28):

$$[1-\beta(1+r)]\delta(1-\tau) - (1+\delta)(1+\beta)(1+r)A^*\big(\delta - [1-\delta(1+r)]\xi^*\big) = 0.$$

Take the total differential with respect to $dr, d\xi$:

$$\begin{aligned} 0 =& \Lambda dr + d\xi \cdot (1+\delta)(1+\beta)(1+r)A^*[1-\delta(1+r)]\xi^* \\ \implies \frac{dr}{d\xi^*} =& \underbrace{\frac{(1+\delta)(1+\beta)(1+r)A^*}{-\Lambda}}_{>0}[1-\delta(1+r)] > 0 \iff \delta(1+r) < 1 \end{aligned}$$

where $\Lambda < 0$ is defined in B.2.2. □

B.2.4 Link between $r - \tau$ in equation (28)

Proposition A.4. *Under Assumption A.3 (ii) and (iii), an increase in the tax rate τ decreases the interest rate according to equation (28).*

Proof. Consider equation (28):

$$[1-\beta(1+r)]\delta(1-\tau) - (1+\delta)(1+\beta)(1+r)A^*\big(\delta - [1-\delta(1+r)]\xi^*\big) = 0.$$

Take the total differential with respect to $dr, d\xi$:

$$\begin{aligned} 0 =& \Lambda dr - d\tau \cdot [1-\beta(1+r)]\delta \\ \implies \frac{dr}{d\xi^*} =& \frac{[1-\beta(1+r)]\delta}{\Lambda} < 0 \end{aligned}$$

where $\Lambda < 0$ is defined in B.2.2. □

C Proofs for the two-instruments case

C.1 Derivations of the FOCs

Consider the Ramsey problem of Section 5.1:

$$\begin{aligned}
\max_{\{\tau_t, r_{t+1}, r^*_{t+1}\}} \quad & \sum_{t=0}^{\infty} \delta^t [u(c_{r,t}) + u(c_{p,t})] \\
\text{s.t.} \quad & c_{r,t} = \frac{1}{1+\beta} 2y_t(1-\tau_t), \\
& c_{p,t} = (1+r_t)2A_{t-1}(\tau_{t-1}), \\
& A_t(\tau_t) = \frac{\beta}{1+\beta} y_t(1-\tau_t), \\
& A_t(\tau_t) + A^*_t(r^*_{t+1}) = (1+r_t)A_{t-1}(\tau_{t-1}) + (1+r^*_t)A^*_{t-1}(r^*_t) - \tau_t y_t.
\end{aligned}$$

Taking the first order conditions I obtain:

$$\begin{aligned}
\tau_t: \quad & 0 = -u'(c_{r,t})\frac{1}{1+\beta}2y_t + 2\delta u'(c_{p,t+1})(1+r_{t+1})\frac{\partial A_t}{\partial \tau_t} + \lambda_t\left[\frac{\partial A_t}{\partial \tau_t} + y_t\right] - \delta\lambda_{t+1}(1+r_{t+1})\frac{\partial A_t}{\partial \tau_t} \\
\implies & 0 = -u'(c_{r,t})\frac{1}{1+\beta}2y_t - 2\delta u'(c_{p,t+1})(1+r_{t+1})\frac{\beta}{1+\beta}y_t + \lambda_t\left[-\frac{\beta}{1+\beta}y_t + y_t\right] + \delta\lambda_{t+1}(1+r_{t+1})\frac{\beta}{1+\beta}y_t \\
\implies & 0 = -u'(c_{r,t})2 - 2\beta\delta u'(c_{p,t+1})(1+r_{t+1}) + \lambda_t + \beta\delta\lambda_{t+1}(1+r_{t+1}) \\
\implies & 0 = [2u'(c_{r,t}) - \lambda_t] + \beta\delta(1+r_{t+1})[2u'(c_{p,t+1}) - \lambda_{t+1}] \\
r_{t+1}: \quad & 0 = u'(c_{p,t+1})2A_t - \lambda_t A_t \\
\implies & 0 = 2u'(c_{p,t+1}) - \lambda_{t+1} \\
r^*_{t+1}: \quad & 0 = \lambda_t \frac{\partial A^*_t}{\partial r^*_{t+1}} - \delta\lambda_{t+1}\left[A^*_t + (1+r^*_{t+1})\frac{\partial A^*_t}{\partial r^*_{t+1}}\right] \\
\implies & 0 = A^*_t\left[\lambda_t\xi^*_t - \delta\lambda_{t+1}\left[1 + (1+r^*_{t+1})\xi^*_t\right]\right]
\end{aligned}$$

that gives the FOCs for the two instrument case:

$$\begin{aligned}
[\tau_t]: \quad & 2u'(c_{r,t}) - \lambda_t + \beta\delta(1+r_{t+1})[2u'(c_{p,t+1}) - \lambda_{t+1}] = 0, \\
[r_{t+1}]: \quad & 2u'(c_{p,t}) = \lambda_t, \\
[r^*_{t+1}]: \quad & \lambda_t\xi^*_t - \delta\lambda_{t+1}\left[1 + (1+r^*_{t+1})\xi^*_t\right] = 0.
\end{aligned}$$

C.2 Various Derivations

Proof of Proposition 2.

Proof. Consider the FOCs above. The second FOC for $t+1$ also gives: $2u'(c_{p,t+1}) = \lambda_{t+1}$. Substituting this in the FOC for τ_t I get $\lambda_t = 2u'(c_{r,t}) = 2u'(c_{p,t})$. Therefore $c_{r,t} = c_{p,t} = C_t$. Using log-utility, I can rewrite the FOC for r^*_{t+1} as:

$$C_t^{-1}\xi^*_t - \delta C_{t+1}^{-1}\left[1 + (1+r^*_{t+1})\xi^*_t\right] = 0.$$

□

RSS minimum level of domestic debt. I show here that the minimum level of domestic debt required to support a full insurance equilibrium is:

$$A_{min} = \frac{\beta}{1+\beta}(1-\tau), \quad \tau \leq 1.$$

Let a_r, a_p be rich and poor agents savings respectively. From individual budget constraints (in the steady state) I get:

$$\begin{aligned} c_r &= 2(1-\tau) + (1+r)a_p - a_r \\ c_p &= (1+r)a_r - a_p \end{aligned}$$

Since full insurance requires $c_r = c_p$, I obtain:

$$a_r = \frac{1}{2+r}2(1-\tau) + a_p$$

Since the borrowing constraint wants $a_r, a_p \geq 0$ and $a_r \geq a_p$ for $\tau \leq 1$, it must be that:

$$a_r \geq \frac{1}{2+r}2(1-\tau) \implies A \geq \frac{1}{2+r}(1-\tau) = \frac{\beta}{1+\beta}(1-\tau).$$

Moreover, if $A = A_{min}$, from the steady-state implementability constraint (30) I get:

$$\begin{aligned} & rA_{min} + r^* A^*(r^*) = \tau \\ & \left(\frac{1}{\beta} - 1\right)\frac{\beta}{1+\beta}(1-\tau) + r^* A^*(r^*) = \tau \\ & \frac{1-\beta}{1+\beta}(1-\tau) + r^* A^*(r^*) = \tau \\ & \frac{1-\beta}{1+\beta} + r^* A^*(r^*) = \tau\frac{2}{1+\beta} \\ & \tau = \frac{1-\beta+(1+\beta)r^* A^*(r^*)}{2}. \end{aligned}$$